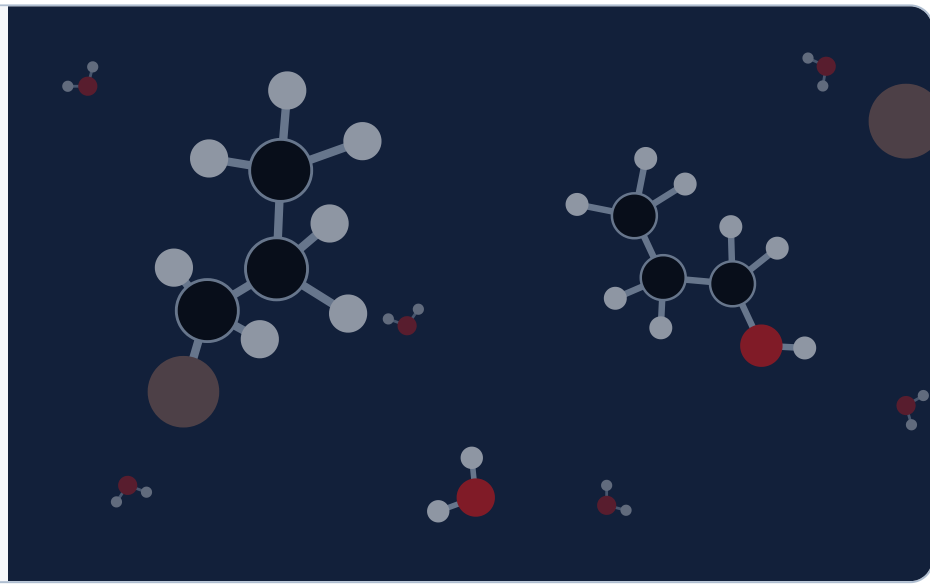
Orgo Helper

Practice naming organic molecules.



DESIGN SYSTEM

Orgo Helper

Brand book, tokens and components · attempt 6 · exported November 6, 2026

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01

Brand book

Orgo Helper is a practice site for organic chemistry students: unlimited practice problems (IUPAC naming, synthesis, more to come), short videos, instant answers. The interface is calm and academic — a white page, a navy frame, one blue for actions — so the molecule is always the most interesting thing on screen. The one playful note is the logo: an O–H bond whose two atoms spell the name, “O·rgo” and “H·elper”, in handwriting, with the oxygen in red as it is in every molecular model kit. Faded ball-and-stick molecules keep the logo company on the navy header and footer, acting out a classic SN2 reaction (1-bromopropane and hydroxide becoming propan-1-ol and bromide), with tiny distant water molecules scattered behind them. That model-kit colour lives only on atoms; the interface itself has no accent colour.

Content fundamentals

- Speak to the student as "you"; the site never says "we" unless it must. "Type a name, get an instant answer, try another."
- Short, plain sentences. Sentence case for headings, buttons and chips ("Start IUPAC practice", "Name this molecule").
- Buttons and links start with a verb: "Start practicing", "Watch the videos", "Next molecule", "Apply disconnection".
- Feedback is encouraging and factual: "Not quite", then the correct name. Never "Wrong!", no exclamation marks (also not in "Retrosynthesis complete" or "Synthesis complete"), no emoji.
- Verdicts are two words or fewer: "Correct", "Not quite", "Answer shown"; a wrong reagent is "Not this one, try again".
- Chemical names are always written exactly (hyphens, commas, locants) and set in **bold** when they are the answer. Reagents use proper subscripts (H_2, $NaBH_4$, PPh_3).
- Settings say what a dependency needs ("Needs Aromatic checked") and how to fix a problem ("Allow it").

Colour

The palette is one light theme, deliberately small:

- Page is `surface`; cards, inputs, panels, dialogs and the molecule panel are `surface-raised`. `surface-sunken` marks a recessed ground inside them (starting-material molecules, hovered rows).
- Header and footer are `navy`. Body text is also `navy`; headings are `heading`; secondary text is `ink-muted`.
- `brand` is the only action colour: primary buttons, links, checked boxes and switches, progress, play buttons, focus. Use it once per view as a button fill. Text on it is `on-brand`.
- **Selected** is `brand-tint` plus `brand` text or a `brand` border, never the tint alone (it is too close to white): the open settings section, a chosen list option, radio card or segment.
- **Current** is `navy`: the current step number in the Stepper.
- `oxygen` (#dc1818) is the logo’s O atom and the O atoms of the decorative molecules in the navy bars, and nothing else: not an accent, never text, a button, link, control or underline anywhere in the interface.

- Atom colours (carbon , hydrogen , oxygen , bromine) appear only on the decorative molecules on navy (MoleculeDeco): 0.55 opacity for the main molecules, 0.35 for the distant water.
- The current Header link is surface-raised (white) with a 3px surface-raised underline. sky is not used in the interface at the moment; it stays in the palette from the earlier logo and the methylamine.
- rule-on-navy is the thin divider above the Footer's legal row.
- Borders: mist for cards, panels and dividers; line-strong for controls a user has to find (inputs, secondary buttons, chips, checkboxes, the switch track).
- error means something to fix: a wrong answer, a wrong reagent, an invalid setting, a conflicting fragment. success is an alias of brand. Both always come with words and an icon (✓, ✕ or !) — and the oxygen red stays on atoms in the navy bars, so a red on the page always means error.
- scrim dims the page behind the settings window, dialogs and phone sheets.
- No gradients. No shadows other than focus-ring, and overlay on things that float above the scrim.

Type

- Source Sans 3 (--font-sans) for everything in the interface. Kalam (--font-hand) for the logo's "rgo" and "elper" letters only. Source Code Pro (--font-mono) for SMILES strings only, never for chemical names.
- One display headline per page at most (home hero). Page and section headings are title ; card titles, dialog titles and feedback headings are subtitle .
- Intro paragraphs use lede in ink-muted ; everything else is body . Labels, nav and chips use label ; fine print and meta use small .
- overline (13px, bold, uppercase, ink-muted) labels parts of a tool: settings section headings, side-list groups, the cells of the plan overview, the panel tag ("Molecule").

Space, shape and layout

- Spacing steps: space-1 4 · space-2 8 · space-3 12 · space-4 16 · space-6 24 · space-8 32 · space-16 64. Cards pad space-6 ; sections are space-16 apart.
- Corners: radius-sm for buttons, inputs, list rows and small cards; radius-lg for cards, panels, dialogs and sheets; radius-pill for chips, badges and round marks.
- Content is capped at content-max (home) or practice-max (practice column), centred, with space-6 side gutters (space-4 on phones). Header is header-height (72px) tall, header-height-phone (64px) on phones; main controls are control-height tall.
- Home page: a centred display headline with one body line in ink-muted under it, then two cards side by side (stacked on phones): **Practice**, holding a TopicList of every generator, and **Learn**, for the videos (a ComingSoonPanel until they exist). The live topic's compact Start button is the page's one brand fill.
- Keyboard focus is always the focus-ring shadow — never remove it.

Generator pages

Every generator page shares one frame: Header, Breadcrumb ("Practice Problems › IUPAC naming"), the title heading with a small "⚙ Settings" button at its right, the generator, then the Footer.

- **IUPAC naming** (one centred column, `practice-max`): settings summary Chips and the problem number, the MoleculePanel, the answer TextField with Submit, Show answer and Skip, a reserved Feedback slot, a primary "Next problem", then the Progress strip.
- **Synthesis** (`oh-gen`: an OptionList `sidebar-width` wide beside the main column): the Stepper, the PlanOverview, then the Workspace (MoleculePanel variant) with MoleculeChips, ReactionArrows, RadioCards for polarity, Callouts and the step's buttons. The side list holds disconnections in stage 1 and reagents in stage 2. Laptop and tablet only on the real site.
- Answers lock once graded: no second attempt, and the primary button becomes the way forward.

Settings

One settings window serves every generator (SettingsWindow): **Molecules**, shared by all generators, then a section per generator. A generator follows the shared Molecules settings unless the student customizes it (RadioCard). Rows are SettingRows using Switch, SegmentedControl, NumberRange, Select and Checkbox grids; rarely used ones hide under a Disclosure; custom fragments are FragmentCards added through a Dialog.

- Changes save automatically and apply to the next problem; there is no Save button.
- Invalid values get an inline `error` message under the field and a "!" on the section in the list; the generator keeps using the last valid settings meanwhile.
- On a first visit to a generator the window opens before the first problem, with a primary "Start practicing".

Phones

- The header (`oh-header--phone`) keeps the logo at 104px, a Sign in or Account link and a 44px menu button.
- Everything tapped is at least `touch-min` (44px). Buttons in a stacked layout go full width; the actions a student needs stay in an ActionBar pinned to the bottom, or on top of the keyboard while typing.
- Windows and side lists become Sheets: settings as a full-height sheet (section list first, then the section with "‹ Settings"), lists as bottom sheets opened from a Picker.
- The Stepper becomes one line with a thin track; the PlanOverview becomes a scrolling strip that expands into a sheet; arrows point down.
- The ⚙ Settings button becomes a 44px icon button.

Components

Components are plain HTML with `oh-` classes from `components/bundle.css` (no JavaScript framework yet). Each component's README gives its markup and rules.

- **Site:** Wordmark, Header (with a phone variant), Footer, Breadcrumb, MoleculeDeco, Card, TopicList, ComingSoonPanel.
- **Actions and forms:** Button (primary, secondary, small, icon, compact, link), Chip, Badge, TextField (answer, locked, SMILES, search), Checkbox, RadioCard, Switch, SegmentedControl, NumberRange, Select (with the phone Picker).
- **Feedback:** Feedback (correct, not quite, answer shown), Progress, Callout; the inline Verdict lives with ReactionArrow.
- **Settings:** SettingsWindow, SettingRow, Disclosure, Dialog, Tabs, Sheet, FragmentCard.
- **Generators:** MoleculePanel (with the Workspace), Stepper, OptionList, PlanOverview, MoleculeChip, ReactionArrow, ActionBar.

Molecules

The molecule drawing and editor are still placeholders: show them inside `MoleculePanel` on `surface-raised`, in `navy` line work, and let the editor colour atoms. Their styling will be designed separately. What the system does fix is the frame around a molecule (MoleculeChip states: starting material, current step, target, preview) and the arrows between molecules (ReactionArrow).

Decorative molecules

- Faded ball-and-stick molecules (MoleculeDeco) sit on the navy bars only, telling one SN2 story: 1-bromopropane beside the Header logo (its chain cropped by the top of the bar) and beside the Footer icon; bromide and propan-1-ol (smaller, further away) beside the Footer's Practice links; water below-right of "Synthesis".
- Tiny distant water molecules (`oh-mol-far` , 0.35 opacity) are scattered evenly across the whole Header and Footer, about one every 90px, never over text and never below the Footer's rule.
- No atom letters, `aria-hidden` , never clickable; model-kit atom colours from the tokens and `mist` bonds; main molecules at 0.55 opacity.
- Page grounds stay plain `surface` : there is no wallpaper, and molecules never decorate cards, panels or the page body.

Logo

- The logo is the Wordmark component (the bromopropane beside it is MoleculeDeco, not part of the logo): inline SVG, coloured by the tokens, so it is never an image file. In the Header it is 122px wide, on phones 104px; `oh-wordmark--lg` (360px) for covers.
- Geometry (116 × 60 units): the O atom (radius 14, `oxygen`) at 14,16; the H atom (radius 9) at 26.67,41.5, so 5.5 units of the bond show between the atoms; "rgo" at 31,23.67 and "elper" at 38.67,50.5, so the H sits on the baseline of "elper" like a capital. Copy the markup from the Wordmark README; don't redraw it.
- The same markup works on light grounds and inside any `oh-on-navy` bar: on light grounds "rgo", "elper" and the H atom are `navy` ; on navy they turn `surface` and the bond turns `mist` . The O is always `oxygen` .
- The icon (`oh-wordmark--icon`) is the same ball-and-stick molecule without letters: on a navy rounded square for favicons, app icons and avatars, and on its own in the Footer.
- Don't recolour the atoms, separate the letters from the atoms, or set the full logo narrower than 96px.

Footer

- The Footer carries the icon (not the full logo; the Header already names the site) with a 1-bromopropane beside it, a one-line tagline in `mist`, link groups (Practice, Learn, Account; bromide, propan-1-ol and water decorate the Practice group) with `overline`-style headings in `surface`, and a legal row (© and Contact, Privacy, Terms) under a `rule-on-navy` divider.
- Tiny distant water molecules are scattered across the footer, above the legal row's rule.
- On phones (`oh-footer--phone`) everything stacks: icon and tagline, the link groups in a two-column grid, then the legal links above ©. Every link is at least `touch-min` tall. The phone footer keeps the bromopropane and a few distant water molecules near the top.

Iconography

Simple line icons, drawn at text size in the colour of the text beside them: chevrons (down, right, back), menu, play, person, settings sliders, search, close ✕, ✓ and the "!" alert. No icon set is chosen yet; no emoji. Skeletal formula fragments drawn in `navy` line work may decorate cards and the hero.

Reference

The two mockup screenshots in `assets/` (home page and practice page) and `assets/Orgo Helper_ Design.md` are the source this system was built from. The settings, IUPAC naming and synthesis generator components come from their lo-fi wireframes.

02

Colour

One light theme, 20 tokens.

surface

`#f5f7fa`

Page background and the practice settings bar. Also the text colour on `navy` (header and footer, and the logo's "rgo" and "elper" letters and H atom), and hydrogen atoms in the decorative molecules (`hydrogen`).

surface-sunken

`#e8edf3`

Recessed grounds inside a raised box: starting-material molecule chips, hovered list rows, the switch thumb's track edge. Text and line work on it stay `navy` (13:1).



heading

`#1d3557`

Headings and card titles on `surface` and `surface-raised` (11.5:1).



brand

`#1d4e89`

The one action colour: primary buttons, links, active card and chip borders, progress fill, play buttons, focus ring. As text on `surface` and `surface-raised` (7.8:1).

brand-tint

`#e4edf7`

The selected state: the chosen settings section, list option, radio card and segment. Always paired with `brand` text or a `brand` border (`brand` on it 7.1:1, `navy` 14:1). Not a second action colour: it never fills a button.

surface-raised

`#ffffff`

Cards, the molecule panel, inputs and feedback boxes. On `navy`: the current Header link and its 3px underline (16:1).



navy

`#12203a`

Header and footer background, all body text on `surface` and `surface-raised` (15:1 and 16:1), and the logo's letters and H atom on light grounds.



ink-muted

`#45526a`

Secondary text: ledes, captions, meta, "Coming later", on `surface` and `surface-raised` (7.3:1).

on-brand → surface-raised

`#ffffff`

Text and icons on a `brand` fill (8.4:1).



sky

`#3386e0`

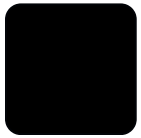
Not used in the interface at the moment (it coloured the earlier logo and the methylamine's nitrogen). Kept for reference; never body text.



oxygen

`#dc1818`

The logo's O atom and the oxygen atoms of the decorative molecules (the Footer's propanol and water, and the distant water across the bars; oxygen is red in molecular models): 5.0:1 on `surface-raised`, 3.2:1 on `navy`. Not an accent colour: never text, a button, link, control or underline, so it can't be confused with `error`.



carbon

`#000000`

Carbon atoms in the decorative molecules (model-kit black), with a 1.2-unit `mist` outline (1 unit on the smaller, further-away propanol) so they read on `navy`. Decoration only, never text.



bromine

`#7f5b52`

Bromine atoms in the decorative molecules (a muted model-kit brown): the bromopropane's bromine and the free bromide in the Footer. Decoration only, never text.



line-strong

`#7d8da6`

Borders of controls a user must find: inputs, secondary buttons, chips (3.1:1 on `surface`, 3.4:1 on `surface-raised`). Also the logo's O–H bond on light grounds.



success → brand

`#1d4e89`

Correct answer — an alias of `brand` for now, so right and wrong differ in hue AND lightness and the palette stays small. Always paired with the word "Correct" and the ✓ icon.



mist

`#aebdd0`

Card borders, dividers and dashed placeholder outlines on `surface`; nav links, the logo's O–H bond and the bonds and carbon outline of the decorative molecules on `navy` (8.5:1); disabled button fill. Decorative only on light grounds (1.8:1) — use `line-strong` for control borders.

hydrogen → surface

`#f5f7fa`

Hydrogen atoms in the decorative molecules (model-kit white).



rule-on-navy

`#2b3d5c`

The 1px divider above the Footer's legal row, on `navy`. Decorative only.



error

`#c01c28`

Something to fix: a wrong answer, a wrong reagent, an invalid setting or a fragment that conflicts with the allowed groups. Feedback border, icon and heading, inline field messages and the "!" marker, on `surface-raised` (6.1:1). Always paired with words and an icon (✕ or !) — and `oxygen` red appears only on atoms (the logo's O and the decorative molecules in the navy bars), never on the page itself.



scrim

`rgba(18, 32, 58, 0.55)`

The dimmed page behind the settings window, dialogs and phone sheets (`navy` at 55%). Click or tap it to close.

03

Type

Family	Stack
`--font-sans`	"Source Sans 3", "Source Sans Pro", system-ui, -apple-system, "Segoe UI", sans-serif
`--font-hand`	Kalam, "Comic Neue", cursive
`--font-mono`	"Source Code Pro", ui-monospace, Menlo, Consolas, monospace

Wordmark

wordmark

`26px / 30px · 700`

rgo elper

The handwritten letters inside the molecule logo ("rgo" on the O, "elper" after the H), set as SVG text on a 116-unit-wide mark. Kalam is used for nothing else.

Interface

display

`56px / 60px · 700 · -0.01em`

Practice naming organic molecules

The one hero headline on the home page.

title

`36px / 44px · 700`

Name this molecule

Page titles and section headings ("Practice topics").

subtitle `24px / 30px · 700`	**IUPAC naming** Card titles and feedback headings ("Not quite").
lede `19px / 30px · 400`	Unlimited IUPAC naming problems, matched to the molecules your course covers. The intro paragraph under a display or title heading.
body `16px / 24px · 400`	Name generated molecules, from simple alkanes to multi-group compounds. Default text, card descriptions, inputs, buttons.
label `14px / 20px · 600`	**Your answer** Form labels, nav links, chips, link-style actions.
small `14px / 20px · 400`	Free. Sign in with Google or Apple. Fine print, meta ("Problem 14"), keyboard hints.
overline `13px / 18px · 700 · 0.06em`	**SYNTHESIS PLAN** Uppercase labels (the component sets text-transform): settings section headings, side-list groups, plan cell labels, the panel tag. Always `ink-muted`.

Code

mono `15px / 22px · 400`	`O=C1CCCCC1` SMILES strings in fragment cards and the SMILES field. Never for chemical names, which stay in `sans`.

04

Spacing, radius and shadow

Spacing

space-1	4px		Icon-to-label gaps.
space-2	8px		Gaps between chips; tight stacks.
space-3	12px		Gaps between buttons; label to input.
space-4	16px		Inner padding of small boxes; gutters on phones.
space-6	24px		Card padding; gaps between cards.
space-8	32px		Heading to content; between blocks in a section.
space-16	64px		Between page sections.

Radius

radius-sm	6px		Buttons, inputs, the keyboard hint.
radius-lg	12px		Cards, the molecule panel, feedback boxes.
radius-pill	999px		Chips and round play buttons.

Shadow

The system uses borders, not shadows. Two exceptions: the focus ring, and `overlay` for things that float above the scrim.

focus-ring	`0 0 0 2px #f5f7fa, 0 0 0 4px #1d4e89`		Keyboard focus on every control: a 2px `surface` gap, then a solid 2px `brand` ring (7.8:1).
overlay	`0 12px 40px rgba(18, 32, 58, 0.28)`		The settings window, dialogs and phone sheets, above the `scrim`. Nothing on the page itself gets a shadow.

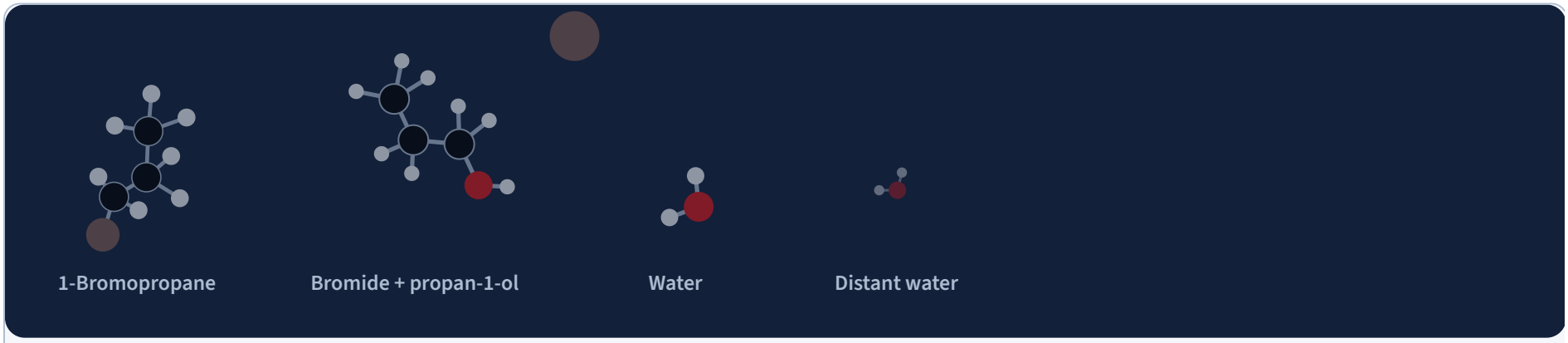
05

Layout

content-max	1216px	Max width of the header, footer and home page content.
practice-max	1016px	Max width of the practice column — narrower so the molecule and answer stay in one glance.
header-height	72px	Height of the navy header bar on laptops and tablets; tall enough for the two-line molecule logo.
header-height-phone	64px	Height of the navy header bar on phones.
control-height	52px	Height of the primary button, answer input, Submit and Skip.
sidebar-width	264px	The side list (disconnections, reagents) beside a generator's workspace.
dialog-width	1120px	Max width of the settings window on laptops; it fills most of the screen on tablets.
dialog-nav-width	248px	The section list on the left of the settings window (about 200px on tablets).
touch-min	44px	Minimum height of anything tapped on a phone: list rows, icon buttons, pickers.

Components: Brand

MoleculeDeco



Faded ball-and-stick molecules that decorate the navy Header and Footer and tell one small story, an SN2 reaction: **1-bromopropane** meets hydroxide and becomes **propan-1-ol**, with **bromide** leaving. Tiny **distant water** molecules are scattered across both bars as solvent in the background. All of it is decoration, never content.

- **1-Bromopropane** (`oh-mol-deco--bromopropane`): in the Header just right of the logo, its bromine sitting by the end of “elper” and its carbon chain rising out of the top of the bar (the bar crops it; that is intended), and in the Footer beside the icon.
- **Bromide + propan-1-ol** (`oh-mol-deco--sn2`): left of the Footer’s Practice group. The bromide is a single small bromine atom; the propanol is drawn smaller, with thinner bonds (`oh-mol-bond--far`) and thinner carbon outlines (`oh-atom--far`), so it reads as further away.
- **Water** (`oh-mol-deco--water`): below-right of the last Practice link.
- **Distant water** (`oh-mol-far`): about one tiny water molecule every 90px across the Header (around the logo, between the logo and the nav, above and below the nav links) and across the Footer (never below the rule above the legal row), each turned a different way. They are 0.22× the water’s size at 0.35 opacity.

- Atoms use model-kit colours from the tokens: `carbon` black with a `mist` outline, `hydrogen` (surface) white, `oxygen` red, `bromine` brown. Bonds are `mist` , round caps. No atom letters: the logo is the only molecule with labels.
- The main molecules are 0.55 opacity (`oh-mol-deco`); the distant water 0.35 (`oh-mol-far`). Everything is `aria-hidden` and ignores the pointer.
- Only on `navy` , never in cards, panels or the page body. The Header and Footer READMEs give the full markup; copy it rather than redrawing.

1-Bromopropane (size and position come from the Header or Footer rules):

```
<span class="oh-mol-deco oh-mol-deco--bromopropane" aria-hidden="tru
e"><svg viewBox="11 -65 92 138" aria-hidden="true"><g class="oh-mol-b
ond"><path d="M36 24 26 58"/><path d="M36 24 22 6"/><path d="M36 24 5
8 36"/><path d="M36 24 65 6.5"/><path d="M65 6.5 87.25 -12.5"/><path
d="M65 6.5 95 25.25"/><path d="M65 6.5 66.75 -34.75"/><path d="M66.75
-34.75 69.5 -68.25"/><path d="M66.75 -34.75 36.75 -39.75"/><path d="M
66.75 -34.75 101 -47"/></g><circle cx="22" cy="6" r="8" class="oh-ato
m oh-atom--h"/><circle cx="58" cy="36" r="8" class="oh-atom oh-atom--
h"/><circle cx="87.25" cy="-12.5" r="8" class="oh-atom oh-atom--h"/><
circle cx="95" cy="25.25" r="8" class="oh-atom oh-atom--h"/><circle c
x="69.5" cy="-68.25" r="8" class="oh-atom oh-atom--h"/><circle cx="3
6.75" cy="-39.75" r="8" class="oh-atom oh-atom--h"/><circle cx="101"
```

```
cy="-47" r="8" class="oh-atom oh-atom--h"/><circle cx="36" cy="24" r
="13" class="oh-atom oh-atom--c"/><circle cx="65" cy="6.5" r="13" cla
ss="oh-atom oh-atom--c"/><circle cx="66.75" cy="-34.75" r="13" class
="oh-atom oh-atom--c"/><circle cx="26" cy="58" r="15" class="oh-atom
oh-atom--br"/></svg></span>
```

Bromide and propan-1-ol:

```
<span class="oh-mol-deco oh-mol-deco--sn2" aria-hidden="true"><svg vi
ewBox="0 0 64 40" aria-hidden="true"><g transform="translate(-5 32) t
ranslate(56 -12) rotate(33) scale(0.5) translate(-56 12)"><g class="o
h-mol-bond oh-mol-bond--far"><path d="M56.0 -12.0 71.08 -20.7"/><path
d="M56.0 -12.0 32.8 -26.5"/><path d="M32.8 -26.5 8.15 -13.45"/><path
d="M8.15 -13.45 -15.05 -27.95"/><path d="M32.8 -26.5 40.05 -48.25"/><
path d="M32.8 -26.5 19.75 -45.35"/><path d="M8.15 -13.45 -3.45 3.95"/
><path d="M8.15 -13.45 18.3 3.95"/><path d="M-15.05 -27.95 -36.8 -19.
25"/><path d="M-15.05 -27.95 -23.75 -49.7"/><path d="M-15.05 -27.95 -
4.9 -49.7"/></g><circle cx="40.05" cy="-48.25" r="4.6" class="oh-atom
oh-atom--h"/><circle cx="19.75" cy="-45.35" r="4.6" class="oh-atom oh
-atom--h"/><circle cx="-3.45" cy="3.95" r="4.6" class="oh-atom oh-ato
m--h"/><circle cx="18.3" cy="3.95" r="4.6" class="oh-atom oh-atom--
h"/><circle cx="-36.8" cy="-19.25" r="4.6" class="oh-atom oh-atom--
h"/><circle cx="-23.75" cy="-49.7" r="4.6" class="oh-atom oh-atom--
h"/><circle cx="-4.9" cy="-49.7" r="4.6" class="oh-atom oh-atom--h"/>
<circle cx="71.08" cy="-20.7" r="4.6" class="oh-atom oh-atom--h"/><ci
rcle cx="-15.05" cy="-27.95" r="9" class="oh-atom oh-atom--c oh-atom-
far"/><circle cx="8.15" cy="-13.45" r="9" class="oh-atom oh-atom--c
oh-atom--far"/><circle cx="32.8" cy="-26.5" r="9" class="oh-atom oh-a
tom--c oh-atom--far"/><circle cx="56.0" cy="-12.0" r="8.5" class="oh-
atom oh-atom--o"/></g><g transform="translate(58 -45) translate(22 2
0) scale(0.5) translate(-22 -20)"><circle cx="22" cy="20" r="15" clas
s="oh-atom oh-atom--br"/></g></svg></span>
```

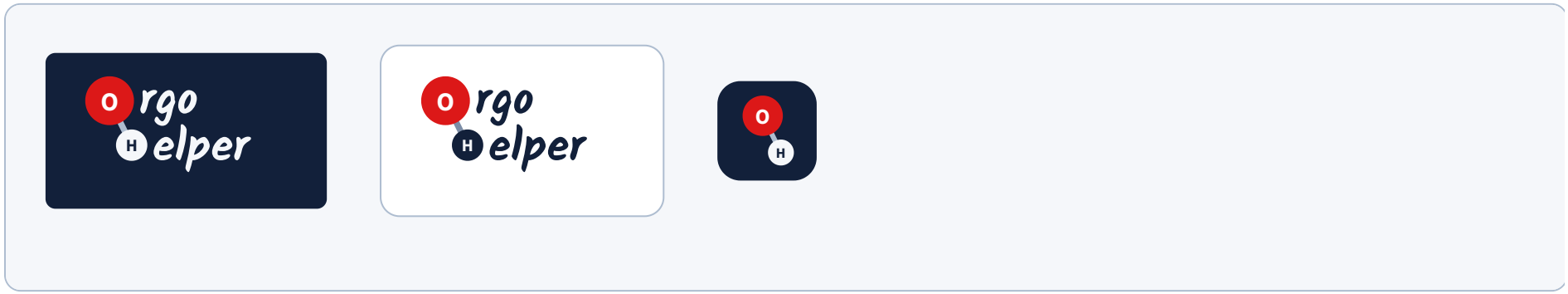
Water:

```
<span class="oh-mol-deco oh-mol-deco--water" aria-hidden="true"><svg
viewBox="0 0 50 48" aria-hidden="true"><g transform="translate(0 0) t
ranslate(34 30) scale(0.5) translate(-34 -30)"><g class="oh-mol-bon
d"><path d="M34 30 10.5 38.6"/><path d="M34 30 31.5 5.1"/></g><circle
cx="10.5" cy="38.6" r="7" class="oh-atom oh-atom--h"/><circle cx="31.
5" cy="5.1" r="7" class="oh-atom oh-atom--h"/><circle cx="34" cy="30"
r="12" class="oh-atom oh-atom--o"/></g></svg></span>
```

One distant water molecule, inside an `oh-mol-far` layer (`translate` places it, `rotat
e` turns it):

```
<g transform="translate(60 22) rotate(20) scale(0.22) translate(-34 -
30)"><g class="oh-mol-bond"><path d="M34 30 10.5 38.6"/><path d="M34
30 31.5 5.1"/></g><circle cx="10.5" cy="38.6" r="7" class="oh-atom oh
-atom--h"/><circle cx="31.5" cy="5.1" r="7" class="oh-atom oh-atom--
h"/><circle cx="34" cy="30" r="12" class="oh-atom oh-atom--o"/></g>
```

Wordmark



The molecule logo: an O–H bond whose atoms spell the name. “rgo” sits on the red (`oxygen`) O atom, “elper” trails the H atom, both in Kalam; 5.5 units of the bond show between the atoms. It is inline SVG coloured by tokens, not an image.

- Markup: `<a class="oh-wordmark" href="/" aria-label="Orgo Helper home">` holding the SVG below (`aria-hidden="true"` on the SVG; the link carries the name). Outside a link use `<span class="oh-wordmark" role="img" aria-label="Orgo Helper">`.
- Colours come from the parent: on light grounds “rgo”, “elper” and the H are `navy` and the bond `line-strong` ; inside `oh-on-navy` the letters and the H turn `surface` and the bond `mist` . The O is always `oxygen` .
- Sizes: 122px wide in the Header (the default), 104px in the phone Header, `oh-wordmark--lg` (360px) for covers and splash. Never below 96px, so the handwritten letters stay legible.
- Icon: `oh-wordmark--icon` with the second SVG below — the same ball-and-stick molecule without letters. On its own it is a 60px navy rounded square for favicons and app icons; inside the Footer it drops the square and shows at 50px (43px on phones).
- Kalam is reserved for the logo’s letters.

```
<svg viewBox="0 0 116 60" aria-hidden="true"><path class="oh-wm-bond" d="M14 16 26.67 41.5"/><circle class="oh-wm-o" cx="14" cy="16" r="14"/><text class="oh-wm-o-t" x="14" y="21.5" text-anchor="middle">O</text><text class="oh-wm-rgo" x="31" y="23.67">rgo</text><circle class="oh-wm-h" cx="26.67" cy="41.5" r="9"/><text class="oh-wm-h-t" x="26.67" y="45" text-anchor="middle">H</text><text class="oh-wm-elper" x="38.67" y="50.5">elper</text></svg>
```

```
<svg viewBox="-6 0 46 52" aria-hidden="true"><path class="oh-wm-bond" d="M14 16 26.67 41.5"/><circle class="oh-wm-o" cx="14" cy="16" r="14"/><text class="oh-wm-o-t" x="14" y="21.5" text-anchor="middle">O</text><circle class="oh-wm-h" cx="26.67" cy="41.5" r="9"/><text class="oh-wm-h-t" x="26.67" y="45" text-anchor="middle">H</text></svg>
```

Components: Layout

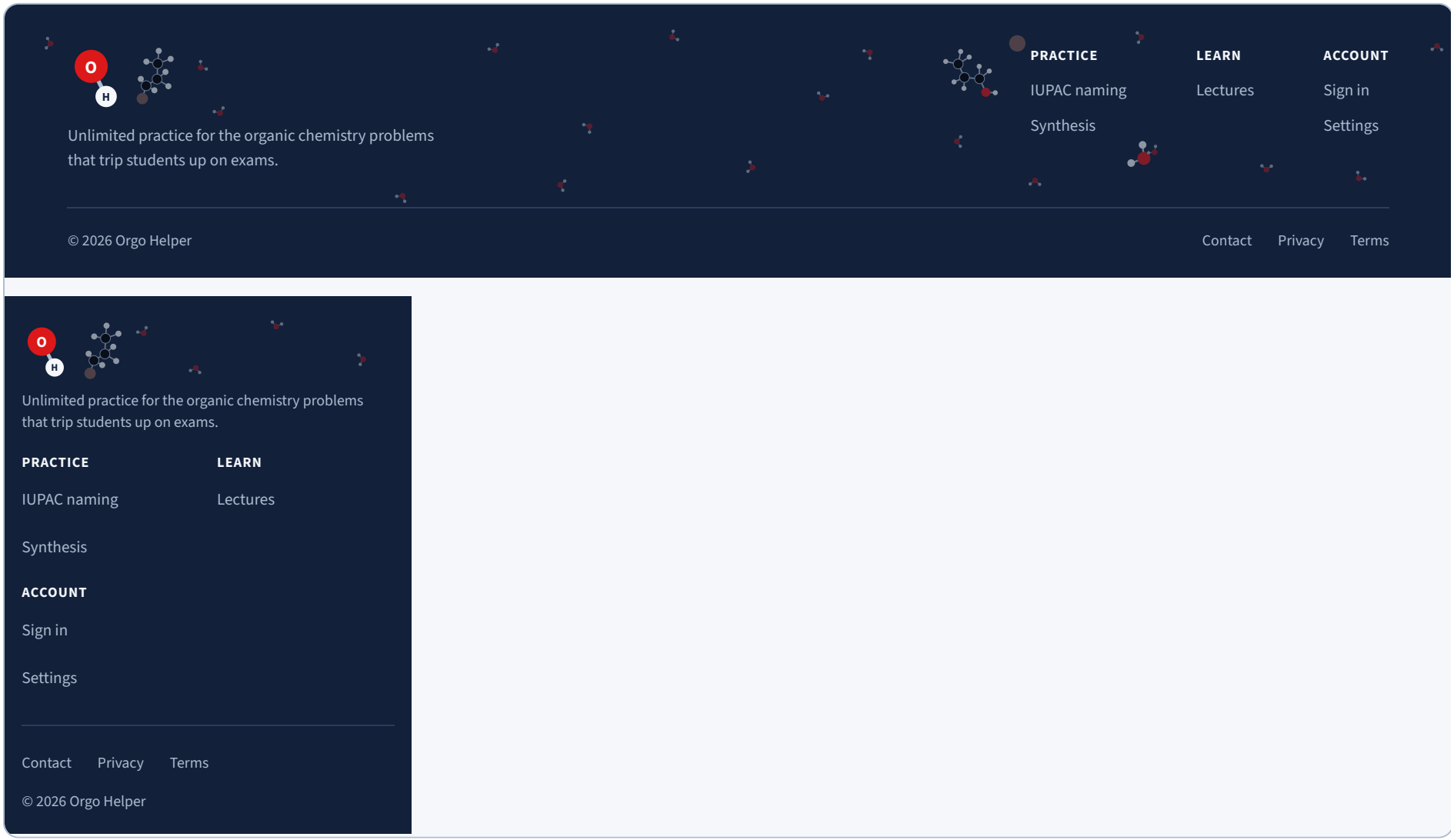
Breadcrumb

Practice Problems › IUPAC naming

The path above a tool page's title, such as "Practice Problems › IUPAC naming".

- Consumer provides the parent links (`oh-link`) and the current page as plain text with `aria-current="page"` ; separate them with a `›` marked `aria-hidden` .
- `oh-breadcrumb` : 14px `ink-muted` , `space-2` gaps. Put it directly above the page title.

Footer



The navy footer: the logo icon with a faded 1-bromopropane beside it, a one-line tagline, link groups with the bromide, propan-1-ol and water molecules by Practice, tiny distant water molecules across the bar, and a legal row. Same bar colour and width as the Header, so the page is framed top and bottom.

- Consumer provides the tagline, the link groups and the legal links. Default groups: **Practice** (one link per generator), **Learn** (Lectures), **Account** (Sign in or Account, Settings); legal row: © year Orgo Helper, then Contact, Privacy, Terms.
- The icon is `oh-wordmark oh-wordmark--icon` inside a home link (`aria-label="Orgo Helper home"`): the Header already shows the full logo, so the Footer repeats only the mark.

- Group headings use the `overline` style in `surface`; links and the tagline are `mist`, turning `surface-raised` with an underline on hover. The legal row sits under a 1px `rule-on-navy` divider.
- **Molecules** (MoleculeDeco, all `aria-hidden`):
- Distant water (`<svg class="oh-mol-far">`, 1440 × 188) comes first inside the footer: 20 molecules spread across the whole bar, around the icon, the tagline, the open middle and the link groups, all above the divider. The legal row stays clear.
- 1-Bromopropane beside the icon, both inside `oh-footer__mark` (32 × 48px).
- Bromide + propan-1-ol (`oh-mol-deco--sn2`) as the first child of `oh-footer__links`, just left of the Practice group (`margin-right: -36px` tucks it in).
- Water inside the Practice group (add `oh-footer__group--water`), placed below-right of the last link.
- **Phones:** add `oh-footer--phone`: everything stacks, the groups form a two-column grid, every link is at least `touch-min` (44px) tall, and the legal links come before ©. Keep the bromopropane beside the icon and a 375 × 200 distant water layer with 4 molecules near the top; leave the bromide, propanol and water out (the grid has no room for them).
- Copy the full markup from `preview.html`; the molecule SVGs are in the MoleculeDeco README.

```
<footer class="oh-footer oh-on-navy">
  <svg class="oh-mol-far" aria-hidden="true" width="1440" height="18
8" viewBox="0 0 1440 188"><!-- distant water molecules --></svg>
  <div class="oh-footer__inner">
    <div class="oh-footer__top">
      <div class="oh-footer__brand">
        <div class="oh-footer__mark">
          <a class="oh-wordmark oh-wordmark--icon" href="/" aria-labe
l="Orgo Helper home"><!-- icon SVG from the Wordmark README --></a>
          <span class="oh-mol-deco oh-mol-deco--bromopropane" aria-hi
dden="true"><!-- 1-bromopropane SVG --></span>
        </div>
        <p class="oh-footer__tagline">Unlimited practice for the orga
nic chemistry problems that trip students up on exams.</p>
      </div>
      <nav class="oh-footer__links" aria-label="Footer">
        <span class="oh-mol-deco oh-mol-deco--sn2" aria-hidden="tru
e"><!-- bromide + propan-1-ol SVG --></span>
        <div class="oh-footer__group oh-footer__group--water"><div cl
ass="oh-footer__heading">Practice</div><a href="#">IUPAC naming</a><a
href="#">Synthesis</a><span class="oh-mol-deco oh-mol-deco--water" ar
ia-hidden="true"><!-- water SVG --></span></div>
        <div class="oh-footer__group"><div class="oh-footer__headin
g">Learn</div><a href="#">Lectures</a></div>
        <div class="oh-footer__group"><div class="oh-footer__headin
g">Account</div><a href="#">Sign in</a><a href="#">Settings</a></div>
      </nav>
    </div>
    <div class="oh-footer__bottom"><span>© 2026 Orgo Helper</span><di
v class="oh-footer__legal"><a href="#">Contact</a><a href="#">Privacy
</a><a href="#">Terms</a></div></div>
  </div>
</footer>
```

Header



The navy site bar: the logo (Wordmark) left, main navigation right, `header-height` (72px) tall, content capped at `content-max`. Faded molecules (MoleculeDeco) sit behind it: 1-bromopropane beside the logo and tiny distant water molecules across the bar.

- Consumer provides the nav links. Mark the current section with `aria-current="page"`: it turns `surface-raised` with a 3px `surface-raised` (white) underline at the bottom of the bar.
- Other links are `mist` and turn `surface-raised` on hover. Keep to three or four top-level items.
- **Phones:** add `oh-header--phone`: `header-height-phone` (64px) tall, the logo at 104px, a Sign in (or Account) link and an `oh-header__menu` button (44px, an inline menu icon, `aria-label="Open menu"`). The menu opens as a full-screen list of Practice Problems, Lectures and Sign in or Account.
- **Molecules:** the bar is `position: relative; overflow: hidden`, so anything that spills past it is cropped.
- The distant water layer (`<svg class="oh-mol-far">`, 1440 × 72; 375 × 64 on phones) comes first inside the header, so the logo and nav paint over it. Laptop: 19 molecules — around the logo, between the bromopropane and the nav, above and below the nav links and at both edges. Phone: 3 — one left of the logo and two between the bromopropane and the Sign in link.
- 1-Bromopropane (`oh-mol-deco--bromopropane`) comes last, as a direct child of the header. The stylesheet places it beside the logo (31 × 47px; 27 × 40px on phones); its chain is cropped by the top of the bar.
- Copy the full markup from `preview.html`; the molecule SVGs are in the MoleculeDeco README.

```
<header class="oh-header oh-on-navy">
  <svg class="oh-mol-far" aria-hidden="true" width="1440" height="72" viewBox="0 0 1440 72"><!-- distant water molecules --></svg>
  <div class="oh-header__inner">
    <a class="oh-wordmark" href="/" aria-label="Orgo Helper home"><!-- logo SVG from the Wordmark README --></a>
    <nav class="oh-nav" aria-label="Main"><a href="#" aria-current="page">Practice Problems</a><a href="#">Lectures</a><a href="#">Account</a></nav>
  </div>
  <span class="oh-mol-deco oh-mol-deco--bromopropane" aria-hidden="true"><!-- 1-bromopropane SVG --></span>
</header>
```

Components: Content

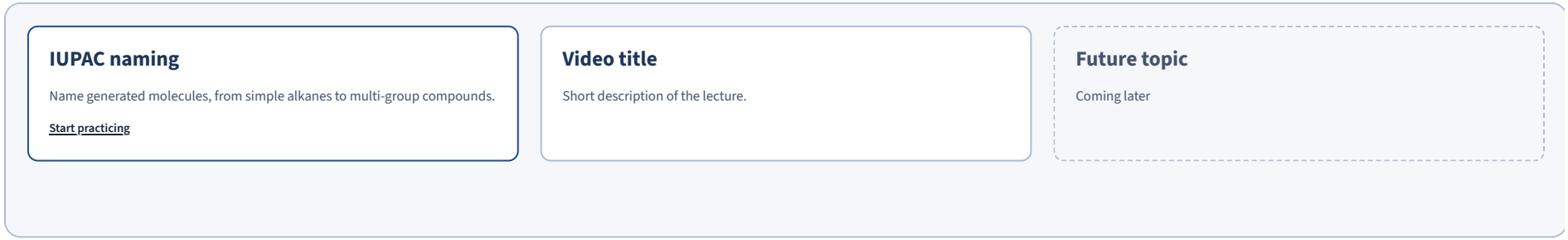
Badge



A small non-interactive pill that states a status, such as "Coming soon".

- Consumer provides the word or two; sentence case, no icon.
- `oh-badge` : `surface-raised` fill, `line-strong` outline, `navy` 14px semibold text. It is not a control: don't make it clickable on its own (the row or card around it can be).
- Use Chip, not Badge, for filters a student can toggle.

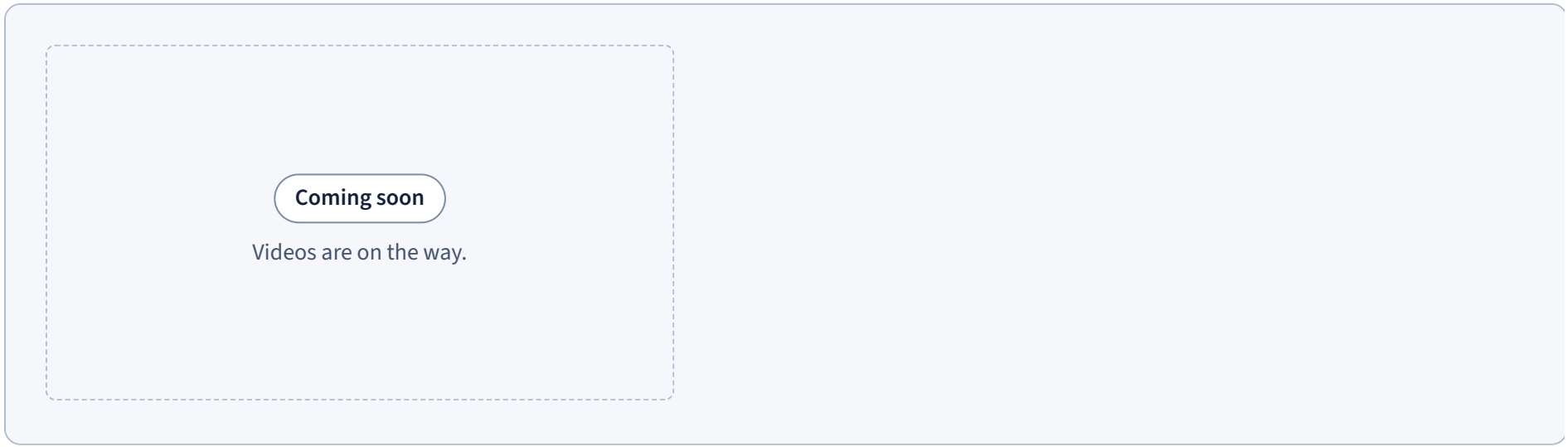
Card



A white box with a `mist` border and `radius-lg` corners, used for practice topics and video tiles.

- Consumer provides a title (`subtitle` style, `heading` colour), a one-line description (`ink-muted`) and at most one link.
- `oh-card--active`: the topic that is open now gets a `brand` border.
- `oh-card--placeholder`: dashed, transparent, for "Coming later" topics.
- Lay cards out three across with `space-6` gaps; stack them on phones.

ComingSoonPanel

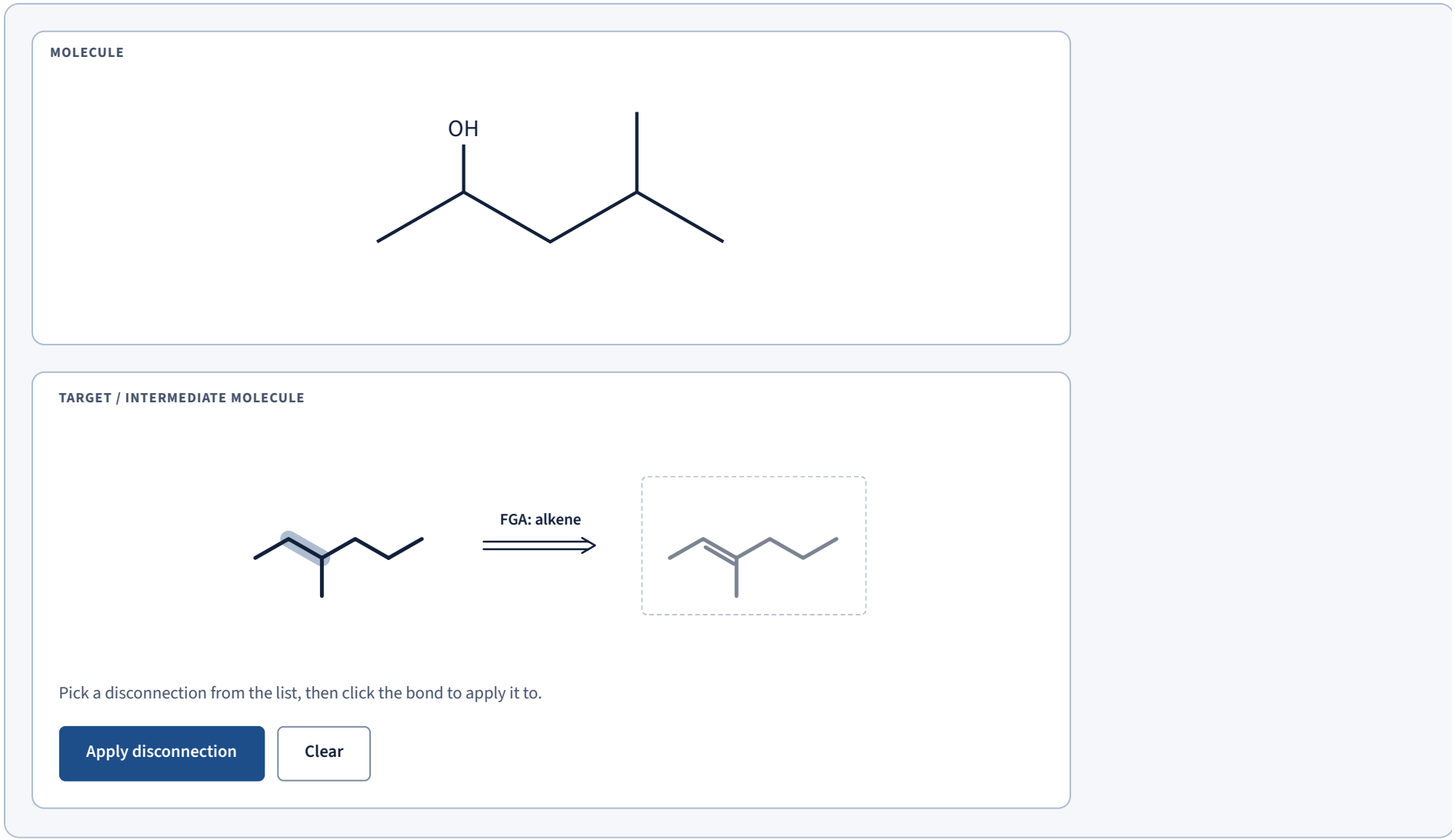


A 16:9 placeholder that stands in for media that doesn't exist yet, such as the videos in the home page's Learn card.

- Consumer provides a Badge ("Coming soon") and one short line ("Videos are on the way.").

- `oh-soon-panel`: `surface` fill, dashed `mist` outline, `radius-sm`; text in `ink-muted`. Replace it with the real thumbnail once the content exists.

MoleculePanel

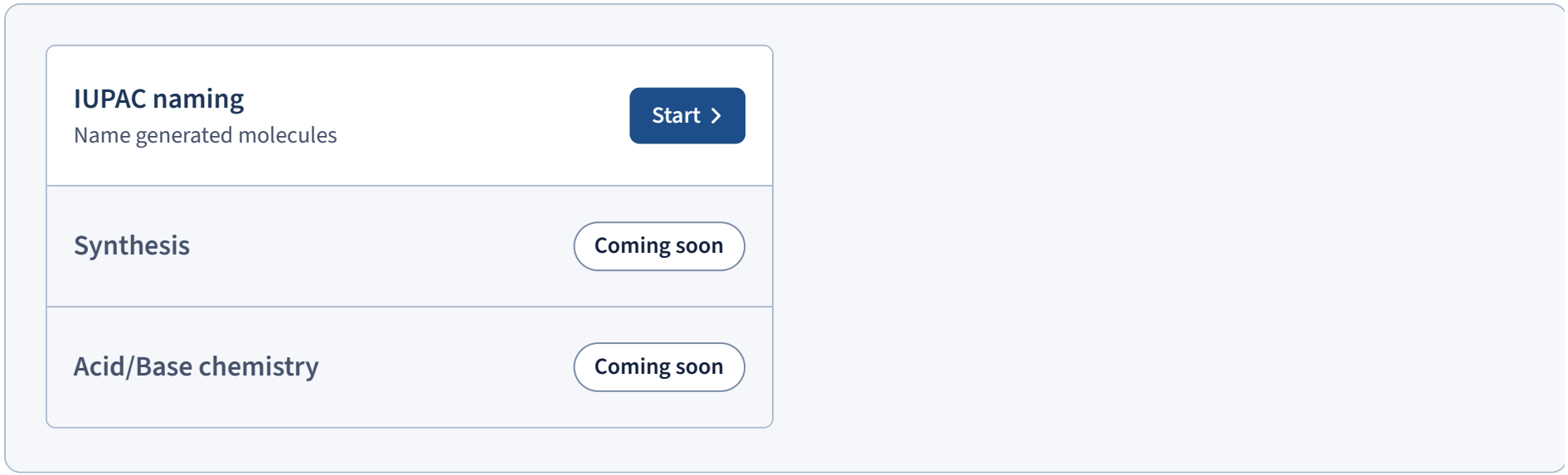


A plain white frame that holds the molecule drawing or the molecular editor, and, as the Workspace, the molecule a student is working on with its arrow, instruction and actions. The molecule styling itself is still a placeholder, to be designed later.

- Consumer provides the drawing (Ketcher or an SVG) inside `oh-molecule`; the frame only sets ground, border, radius and centring. An optional `oh-molecule__label` (`overline`, "Molecule", "Preview") tags it at the top left.
- **Workspace** (`oh-workspace`): label, then `oh-workspace__stage` (molecules, ReactionArrows, RadioCards or a Callout, centred in a row), then `oh-workspace__instr` (one `ink-muted` sentence: "Pick a disconnection from the list, then click the bond to apply it to."), then the button row. `oh-workspace--phone` stacks the stage vertically with down arrows.

- In the IUPAC generator the panel is 340px tall on laptops and shrinks (220 → 150px) on phones when the keyboard opens.
- Line work is `navy`; atom colours come from the editor for now. Don't add brand colour to bonds or atoms. A selected bond is highlighted with a wide `mist` stroke behind it.
- Ketcher, when embedded (the fragment dialog's Draw tab), keeps its own toolbars inside this frame; load it only when that view opens.

TopicList

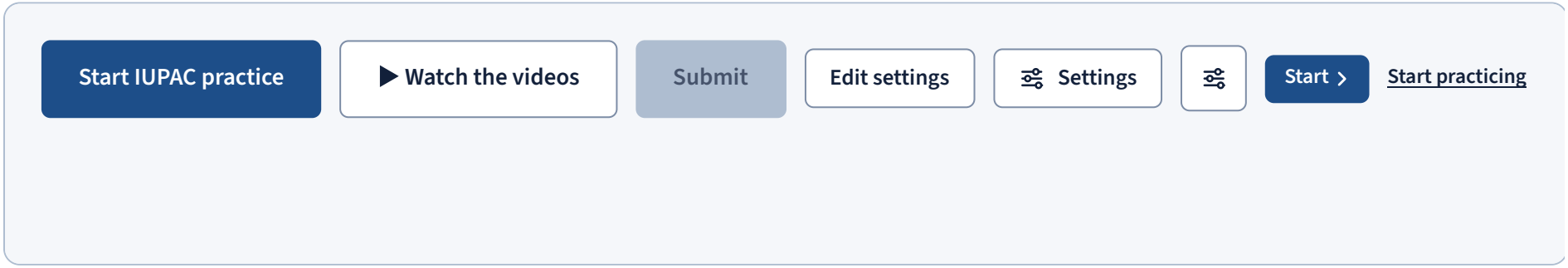


The list of practice generators, one row each, used in the home page's Practice card.

- Consumer provides one row per generator: its name, and for a live generator a one-line description and its link.
- Live topic (`oh-topic oh-topic--live`): raised row, name in `heading` , description in `ink-muted` , and a compact primary Start button (`oh-btn oh-btn --primary oh-btn--compact`) at the right. It is the page's one `brand` fill.
- Coming-soon topic (`oh-topic`): `surface` row, name in `ink-muted` , a Badge reading "Coming soon" at the right; it links to that generator's Coming-soon page.
- The whole row is the link and the tap target; the Start button inside it is a `<span>` , not a second link.
- Live topics first, then coming-soon topics. Rows are at least 56px tall (68px live).

Components: Actions

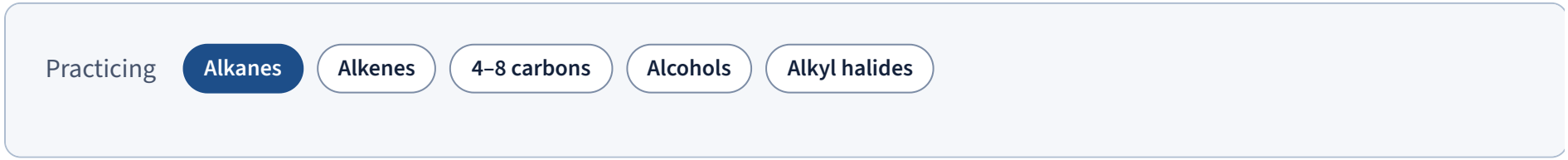
Button



One primary action per view in `brand`; everything else is the outlined secondary button or a text link.

- **Primary** (`oh-btn oh-btn--primary`): the thing the page is for — "Start IUPAC practice", "Submit", "Next molecule".
- **Secondary** (`oh-btn`): white with a `line-strong` border — "Watch the videos", "Skip", "Show answer".
- **Small** (`oh-btn--small`): toolbars and dialogs, such as "⚙ Settings" beside a generator's title, and Cancel / Add fragment.
- **Icon** (`oh-btn oh-btn--icon`): 44px square with only an icon, such as ⚙ Settings beside the title on phones. Always give it an `aria-label`.
- **Disabled**: the `disabled` attribute; `mist` fill, `ink-muted` label. Submit stays disabled until an answer is typed; Submit, Show answer and Skip disable once a problem is answered; Next step stays disabled until the right reagent is picked.
- **Link** (`oh-link`): underlined, for in-card actions ("Start practicing", "Edit shared", "Restart").
- Labels are verb-first, sentence case. An icon may lead the label (the settings sliders for Settings).
- **Compact** (`oh-btn oh-btn--primary oh-btn--compact`): 32px, for an action inside a row, such as Start on the live topic in a TopicList. The row stays the tap target, so it remains at least 44px tall on phones.

Chip

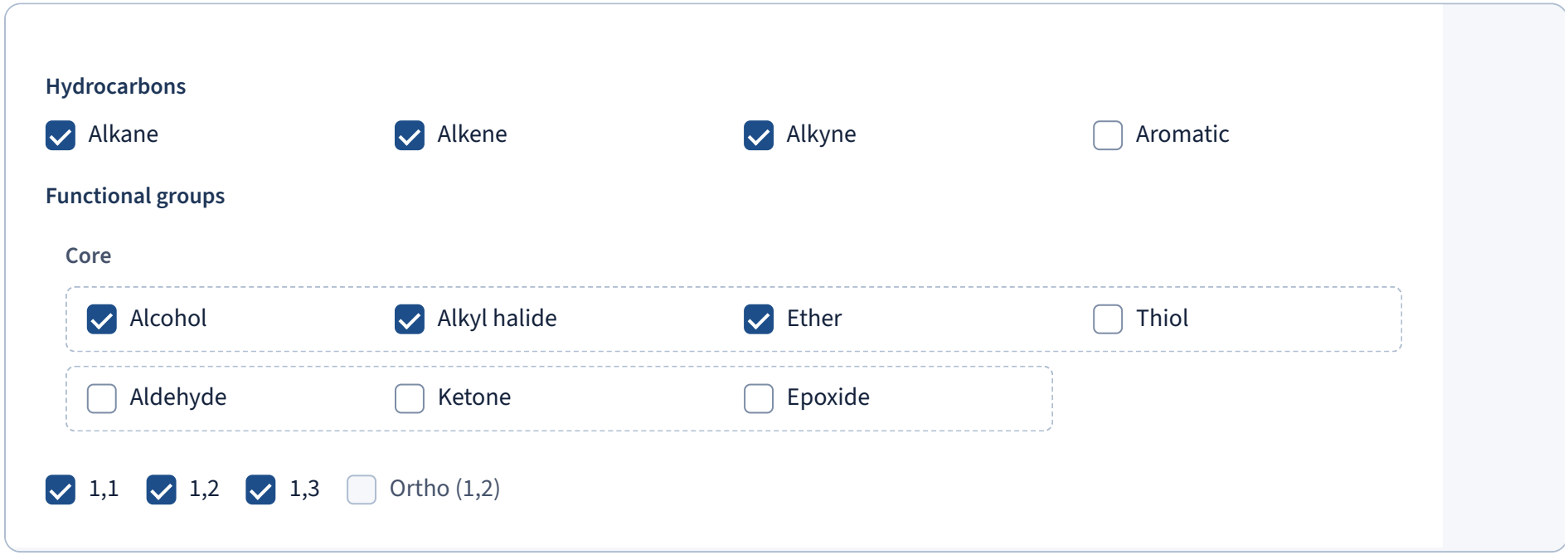


A pill showing one practice filter (a functional group, a carbon range) in the settings bar.

- Consumer provides the label and, if it can be toggled, `aria-pressed`. Pressed chips fill with `brand`.
- Lead a row of chips with a "Practicing" label in `ink-muted`. Labels are short nouns, sentence case.
- **Settings summary**: above a problem, unpressed chips can summarize what's being practiced ("Alkane, alkene, alkyne"); clicking one opens the settings window. On phones the row scrolls sideways.

Components: Forms

Checkbox



A square check for each item the student can allow or exclude, laid out alone, in a row, or as a grid with group headings.

- Markup: `<label class="oh-check"><input type="checkbox"> Alkane</label>`. Always a real `<input>` inside its `<label>`; checked boxes fill with `brand` and a white tick.
- **Grid** (`oh-checkgrid`): four columns on laptops, two on phones (`oh-checkgrid--2`, or automatically below 560px). `oh-checkgrid__head` names a group ("Hydrocarbons", "Functional groups"); `oh-checkgrid__head--sub` names an indented subgroup ("Core", "Carboxylic acid derivatives").
- **Cluster** (`oh-checkgrid__cluster`): a dashed `mist` outline around groups that belong together, such as alcohol, alkyl halide, ether and thiol in Core. `--partial` spans three columns.
- **Inline** (`oh-checkgrid--inline`): a short row inside a setting row (ring topology, 1,1–1,5 group distance).
- **Disabled**: `disabled` on the input; label turns `ink-muted`. Say why in the row's description ("Needs Aromatic checked").
- A group heading may carry a "Select all · Clear" pair of `oh-link`s on the right.

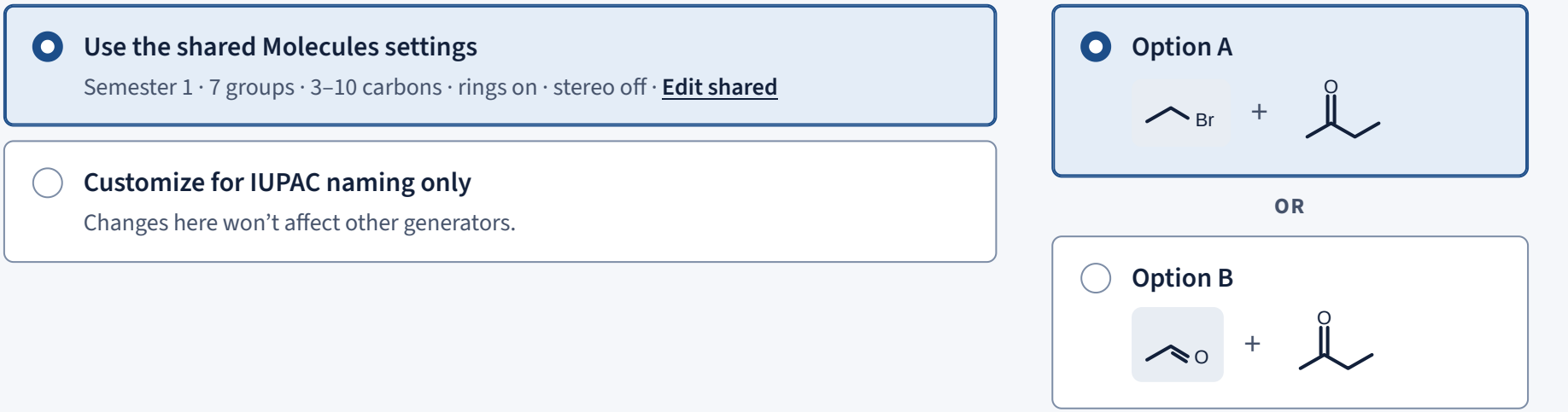
NumberRange



A minimum and maximum pair of small number inputs joined by "to" ("Molecule size: 3 to 10").

- Markup: `<span class="oh-range"><input class="oh-num" type="number" aria-label="Minimum carbons"> to <input class="oh-num" type="number" aria-label="Maximum carbons"></span>`. A single `oh-num` works for one limit ("Starting materials per problem").
- Each input gets its own `aria-label`; the setting row's label names the pair.
- Invalid: `aria-invalid="true"` on the input at fault gives it a 2px `error` border; the row shows an `oh-field-msg` ("Minimum is larger than maximum.").
- Disabled (`disabled`): `surface` ground, `mist` border, for limits that depend on another setting (amine counts need Amine checked).

RadioCard



A bordered card with a radio button, title and description, for choosing one of two or three options that each need a sentence.

- Used for "Use the shared Molecules settings" / "Customize for IUPAC naming only", and for the two polarity options of a disconnection (each holding its molecules).
- Markup: `<label class="oh-radio-card"><input type="radio" name="…"><span class="oh-radio-card__text"><span class="oh-radio-card__title">…</span><span class="oh-radio-card__desc">…</span></span></label>` inside an `oh-radio-cards` wrapper (`role="radiogroup"` with a label).
- Selected: `brand` ring, `brand` border and `brand-tint` ground. Never `brand` as a solid fill.
- The description can hold a summary of the current values and an `oh-link` ("Edit shared").
- Expanded content (the customized copy of Molecules) goes in `oh-radio-card__more` under the selected card, not inside the label.
- Between options that read as alternatives, an `oh-or` separator ("Or") may sit between cards.

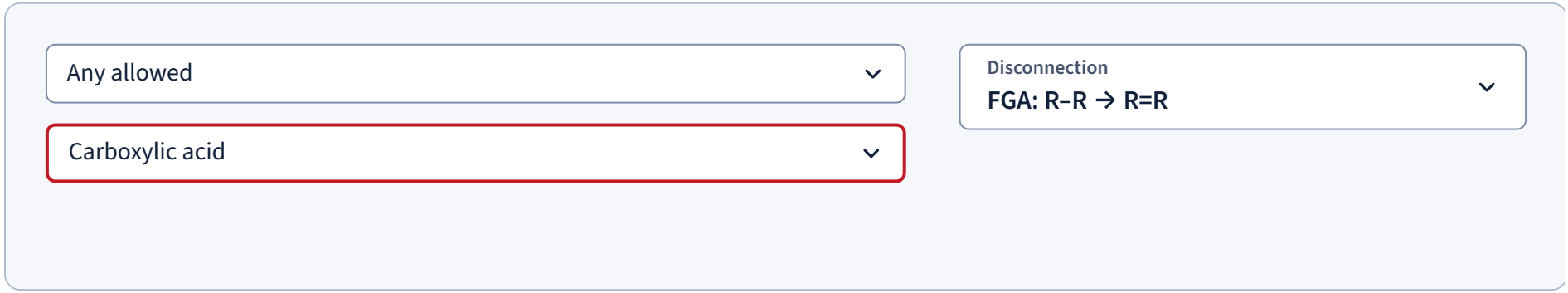
SegmentedControl



Two or three mutually exclusive values shown side by side ("Off · Include", "Forward · Retrosynthesis · Mix").

- Markup: `<div class="oh-seg" role="group" aria-label="…">` holding `<button aria-pressed="true|false">` per value. Exactly one is pressed.
- Pressed: `brand-tint` ground, `brand` text and a 2px inset `brand` border. Labels are one or two words, sentence case.
- More than three values, or long labels: use a Select.

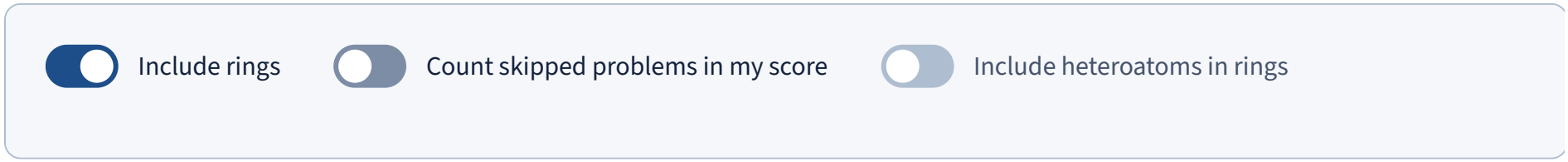
Select



A dropdown for one value from a longer list ("Most senior functional group"), and its phone form, the Picker.

- **Select** (`oh-select`): a native `<select>` restyled, 36px tall inside setting rows, with a navy chevron. Invalid: `aria-invalid="true"` and an `oh-field-msg` under the row.
- **Picker** (`oh-picker`): a full-width button on phones that shows a small caption (`oh-picker__cap` , "Disconnection") over the current value (`oh-picker__val`) and opens a bottom Sheet with the full list. It replaces a side list (OptionList) on phones. `aria-haspopup="dialog"` .

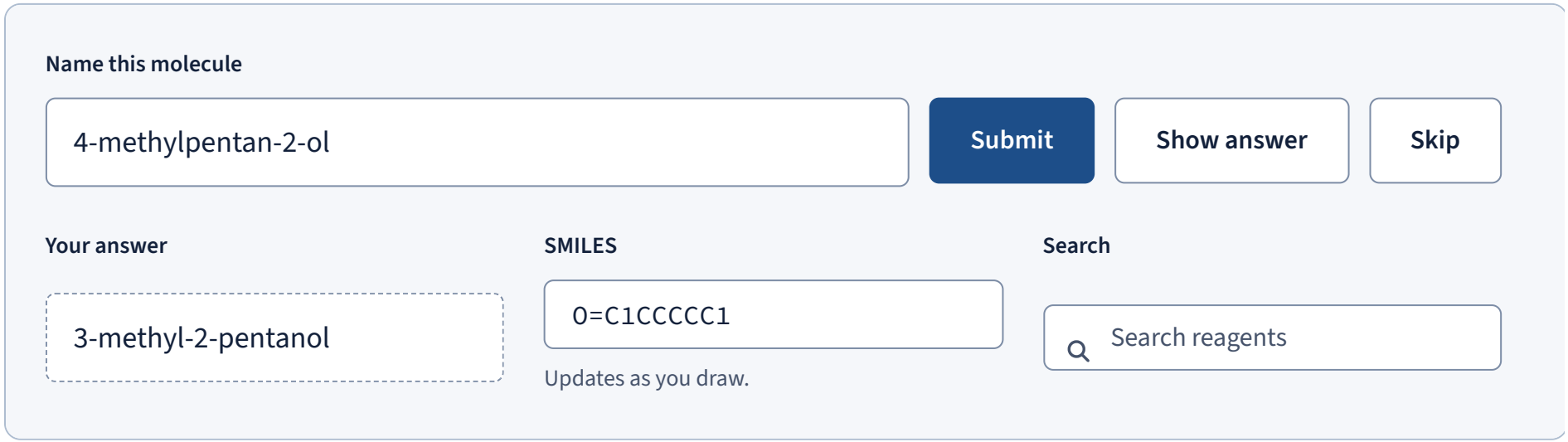
Switch



An on/off setting that takes effect without a Save button ("Include rings", "Keyboard shortcuts").

- Markup: `<input type="checkbox" role="switch" class="oh-switch" aria-labelledby="…">`, labelled by its setting row's label.
- Off: `line-strong` track (3.4:1). On: `brand` track. The white thumb moves right.
- Use a Switch for a single yes/no; use a SegmentedControl when the two values need words ("Off · Include"), and a Checkbox for picking items from a list.

TextField

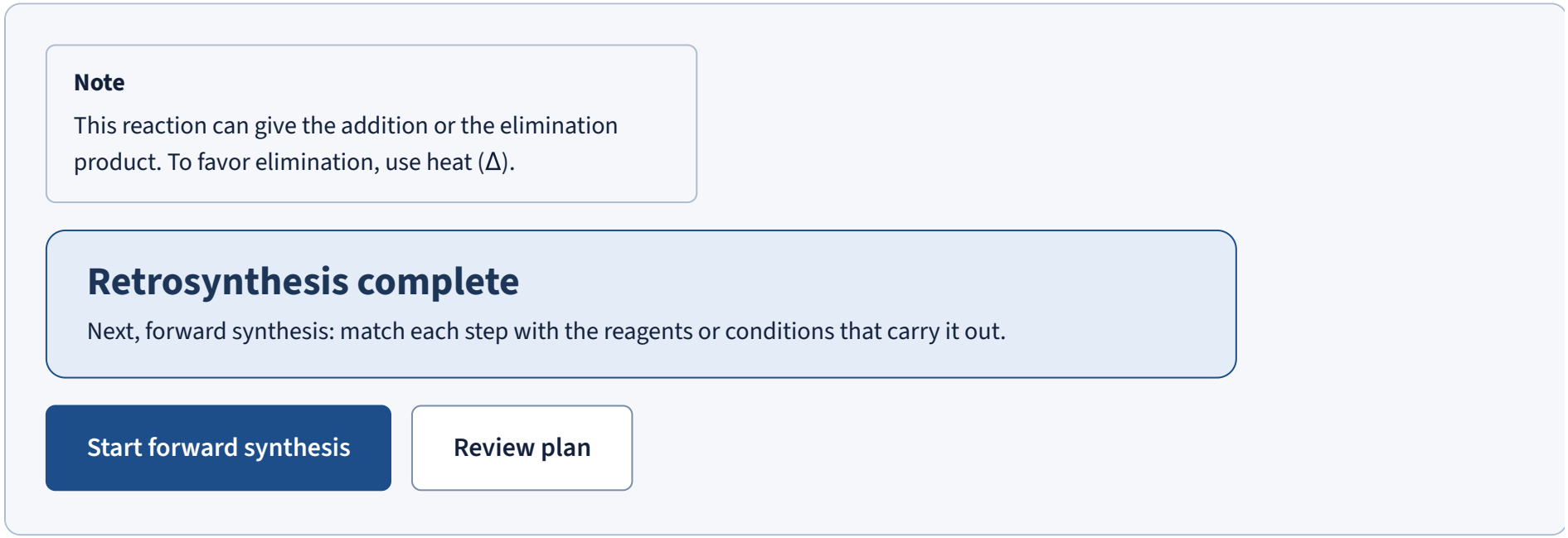


A text input with a visible label above it: the answer box, the SMILES field and search.

- Consumer provides the label, the input's value and the buttons. Always a real `<label>`; never placeholder-only (search inside a list is the exception: it gets an `aria-label`).
- **Answer** (`oh-input`, `control-height`): 18px text so chemical names (hyphens, locants) are easy to proofread. Pressing Enter submits. It has focus when a problem loads. Turn off autocorrect, autocapitalize and spellcheck. Actions sit on the same row on laptops (Submit, Show answer, Skip) and stack below it on phones.
- **Locked** (`readonly`): after Submit or Show answer the answer stays visible with a dashed border and can't be edited.
- **SMILES** (`oh-input--mono`): `mono` type, for the fragment dialog.
- **Small** (`oh-input--sm`): 40px, inside dialogs and lists.
- **Search** (`oh-search` wrapper with a leading search icon): at the top of long lists such as reagents.
- **Hint** (`oh-field__hint`) under the input; **invalid** `aria-invalid="true"` plus an `oh-field-msg`.

Components: Feedback

Callout



A short boxed message in the workspace: a **note** beside a step that needs explaining, or the **done** message at the end of a stage.

- Note (`oh-callout`): `surface` ground, `mist` border, a bold lead word ("Note") and one or two sentences ("This reaction can give the addition or the elimination product. To favor elimination, use heat (Δ)."). Beside the product on laptops, below it on phones.
- Done (`oh-callout--done`): `brand-tint` ground and `brand` border, a `subtitle` heading that states the result ("Retrosynthesis complete") and one sentence on what comes next. The primary button that follows ("Start forward synthesis") is the view's one `brand` fill.
- Tone as everywhere: factual and encouraging, no exclamation marks. Not for answer feedback: that is Feedback or a Verdict.

Feedback

Not quite

Correct name **2-chloro-4-methylpentane**

Your answer 2-chloro-3-methylpentane

Correct

Your answer 3-methyl-2-pentanol

Preferred name **3-methylpentan-2-ol**

Synonyms 3-methyl-2-pentanol (your answer)

Answer shown

Correct name **4-methylpentan-2-one**

Synonyms 4-methyl-2-pentanone
methyl isobutyl ketone

Your attempt 4-methylpentan- (not graded)

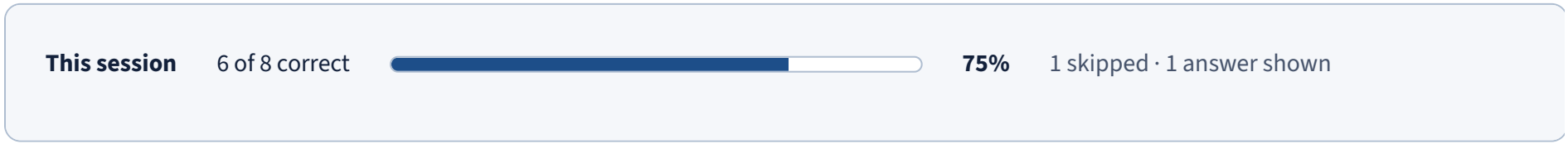
Press Enter for the next molecule

The box that appears under the answer after Submit or Show answer: icon, a short verdict, the names, and what the student typed.

- Consumer provides the verdict, correct name(s) and the student's answer. Use `role="status"` so it is announced.
- **Correct** (default): `success` colour, ✓, heading "Correct". When the student used an accepted older form, show both: "Your answer" and "Preferred name", then any synonyms, tagging the one they typed "(your answer)".
- **Wrong** (`oh-feedback--wrong`): `error` colour, ✕, heading "Not quite". "Correct name" (bold) first, then "Your answer".

- **Answer shown** (`oh-feedback--shown`): neutral, dashed `line-strong` border and a dashed "?" mark, heading "Answer shown". Correct name, synonyms, then "Your attempt" tagged "(not graded)".
- Detail rows: `<dl class="oh-feedback__rows">` with `dt` labels in `ink-muted` and `dd` values; a synonyms `dd` holds a plain `ul` . On phones the labels stack above the values. Leave the synonyms row out when there are none.
- Reserve the space (`oh-feedback-slot` , empty) before an answer so Next doesn't jump down the page.
- Follow it with a primary "Next problem" button and the Enter hint (`oh-kbd`). For a single step inside a workspace, use the inline Verdict (ReactionArrow) instead.
- Colour is never the only signal. Tone: encouraging and factual — no exclamation marks, no "Wrong!".

Progress

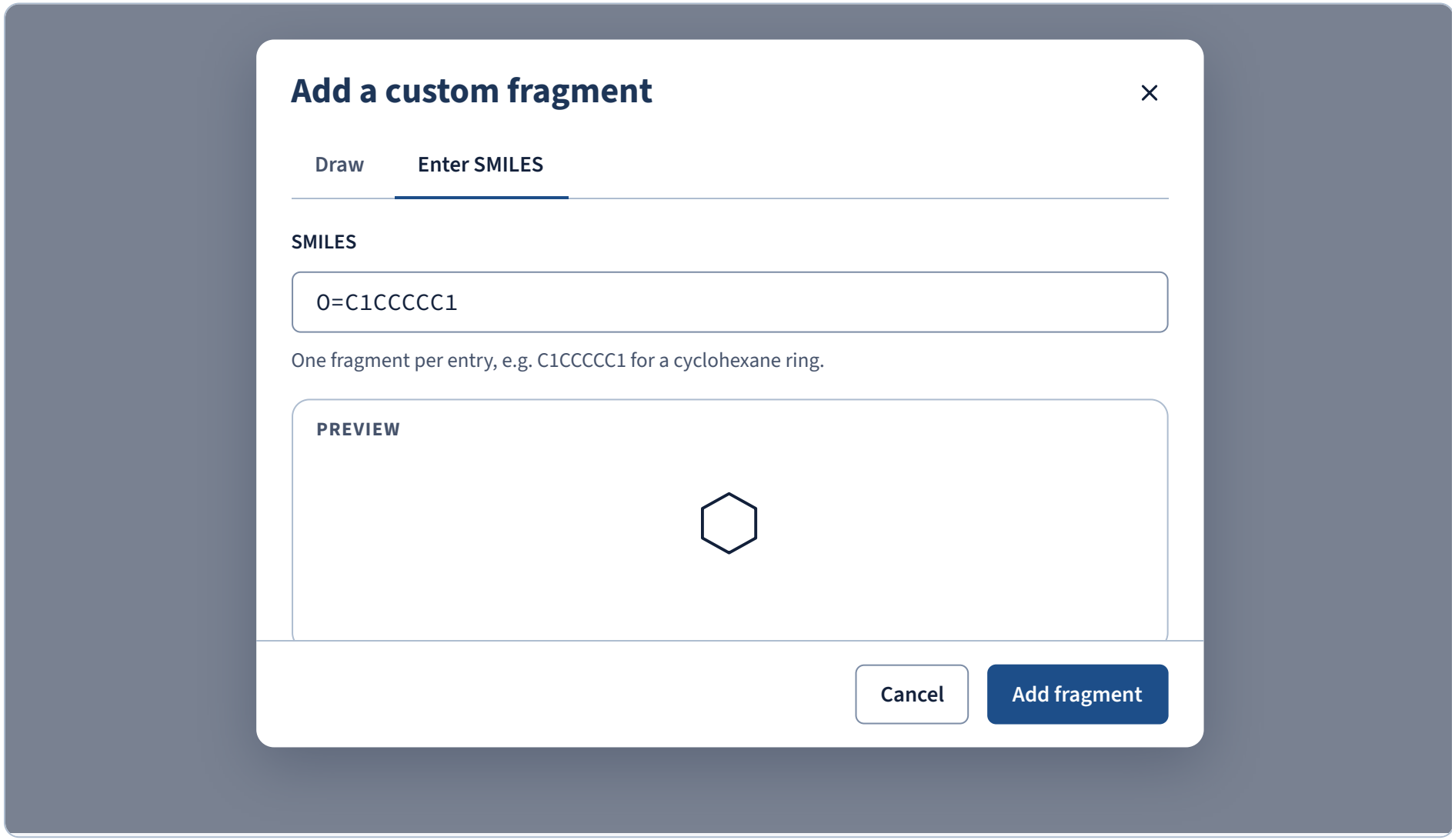


The session score strip at the bottom of the practice column: counts, a `brand` bar, the percentage, skips and answers shown.

- Consumer provides correct, attempted, skipped and shown counts. Always show the numbers as text beside the bar. Shown answers don't count toward the percentage; skipped ones count only if the student turns that on in Settings.
- It wraps onto two lines on narrow screens. On phones the compact score ("Score 5/6 · 83%") sits beside the problem number at the top, and the full strip stays below the fold.
- The same track, thinner, is the stage progress in the phone Stepper.

Components: Settings

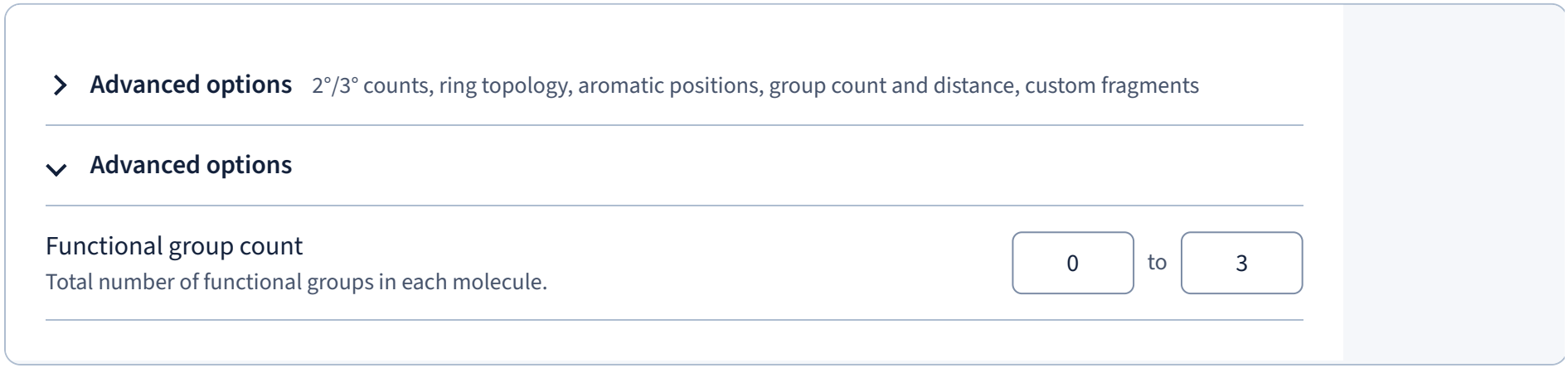
Dialog



A box that floats over the dimmed page for a short, focused task, such as adding a custom fragment.

- Markup: an `oh-scrim` behind an `oh-dialog` (`role="dialog" aria-modal="true" aria-labelledby` its title). Parts: `oh-dialog__head` (title in `subtitle` style and an `oh-close` ✕ button with `aria-label="Close"`), optional Tabs, `oh-dialog__body`, `oh-dialog__foot`.
- Ground `surface-raised`, `radius-lg`, the `overlay` shadow over `scrim`. It is the only place besides Sheet that uses a shadow.
- Footer: secondary Cancel then primary action, right-aligned (`oh-dialog__foot--end`), both `oh-btn--small`. Or, for settings, the `oh-saved` status on the left and a reset link on the right.
- Close with ✕, Esc or a click on the scrim; focus returns to the button that opened it. A dialog can open over the settings window (its scrim dims the window too).
- Phones: a Dialog becomes a full-height Sheet.

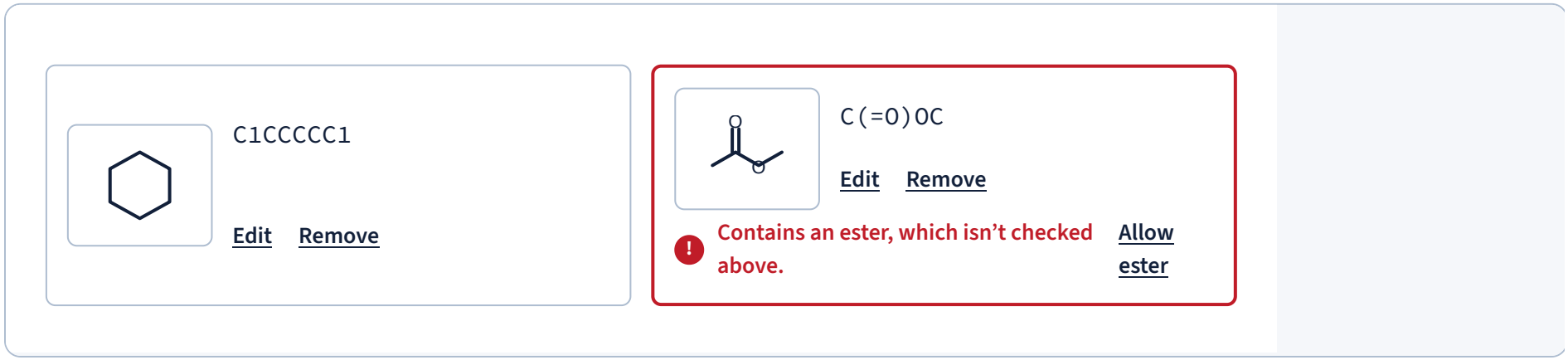
Disclosure



A row that hides less-used settings until it is opened ("Advanced options").

- Markup: `<details class="oh-disclosure"><summary>Advanced options <span class="oh-disclosure__hint">2°/3° counts, ring topology, …</span></summary> … rows … </details>`.
- The hint lists what is inside, so students know whether to open it. The chevron points right when closed and down when open.
- Start closed. On phones, open Advanced options as its own page in the Sheet (with a back button) rather than expanding inline.

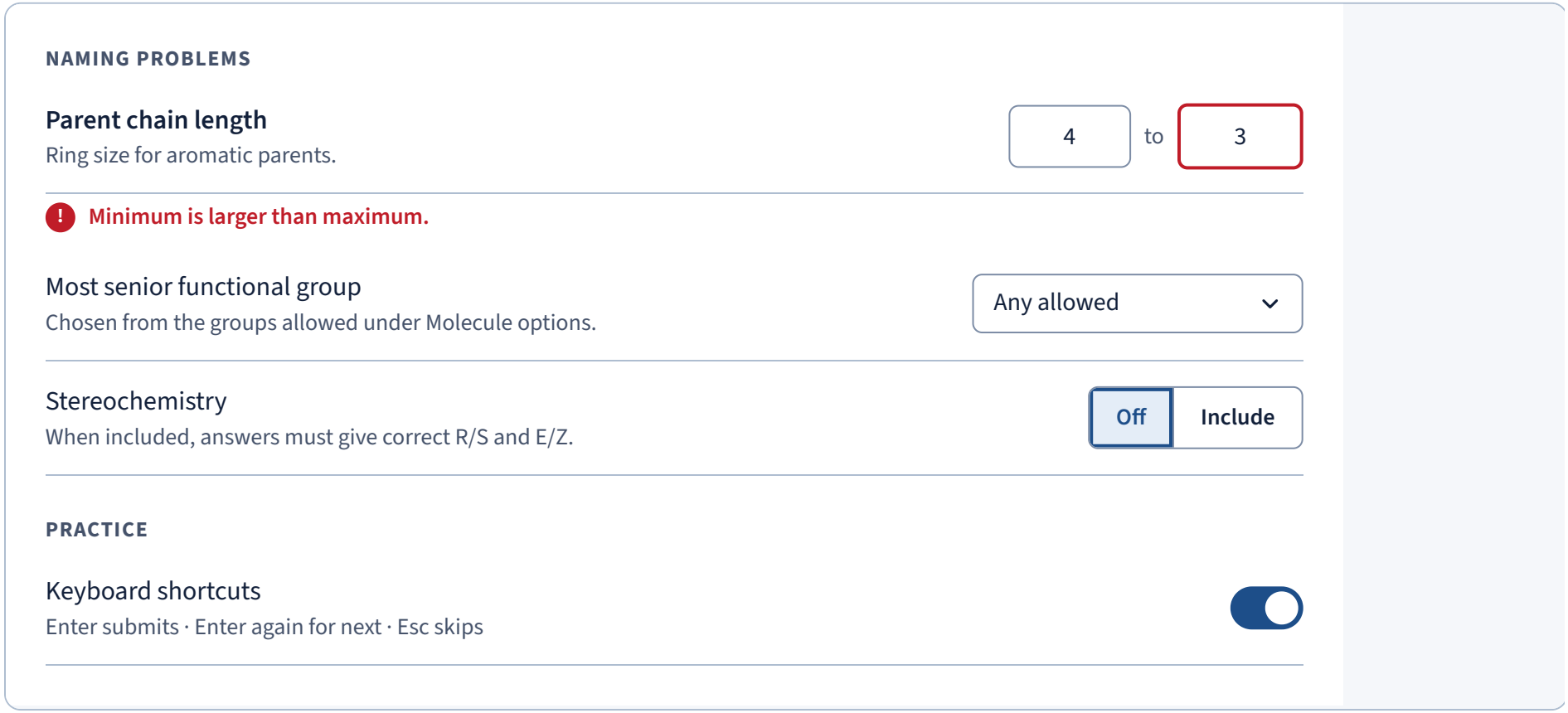
FragmentCard



A custom fragment every generated molecule must contain: its drawing, its SMILES, and Edit / Remove.

- Markup: `<div class="oh-frag"><span class="oh-frag__pic">…svg…</span><span class="oh-frag__smiles">C1CCCCC1</span><span class="oh-frag__acts"><button class="oh-link">Edit</button><button class="oh-link">Remove</button></span></div>`, in an `oh-frags` grid (two across in the settings window, one on phones) under a block SettingRow with a small "+ Add fragment" button.
- SMILES is set in `mono`. The drawing follows the MoleculePanel rule: `navy` line work, colour left to the editor.
- **Conflict**: `oh-frag--invalid` (2px `error` border) plus an `oh-field-msg` inside the card with its one-click fix ("Contains an ester, which isn't checked above. Allow ester").
- "+ Add fragment" and Edit open the fragment Dialog (Draw in Ketcher, or Enter SMILES).

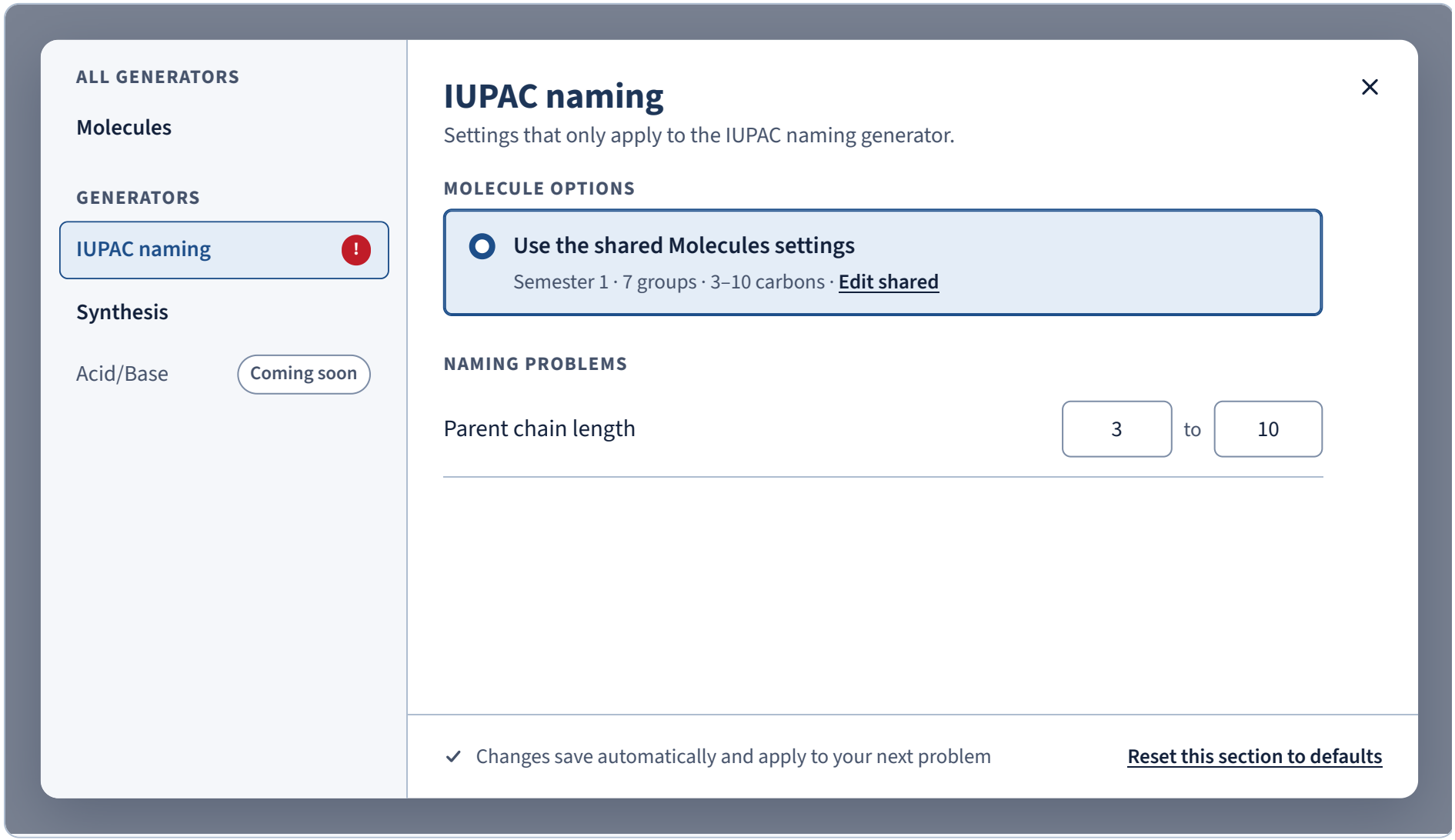
SettingRow



One setting: its name and an optional one-line description on the left, its control on the right, a `mist` rule below.

- Markup: `<div class="oh-setting"><div class="oh-setting__text"><span class="oh-setting__label" id="…">Molecule size</span><span class="oh-setting__desc">Number of carbons in each molecule.</span></div><div class="oh-setting__control">…</div></div>`.
- Controls: Switch, SegmentedControl, NumberRange, Select, an inline Checkbox row, or an `oh-link` ("Edit").
- `oh-setting--block` stacks the control under the text, for a checkbox grid or fragment cards.
- Group rows under an `oh-section-label` (uppercase `overline`: "Naming problems", "Practice").
- **Invalid**: add `oh-setting--invalid` (label turns semibold), mark the control `aria-invalid="true"`, and follow the row with `<p class="oh-field-msg">` naming the problem and, when there is one, a one-click fix as an `oh-link` ("Allow it"). Point the control's `aria-describedby` at the message.
- Descriptions are plain sentences; say what a dependency needs ("Needs Aromatic checked").

SettingsWindow

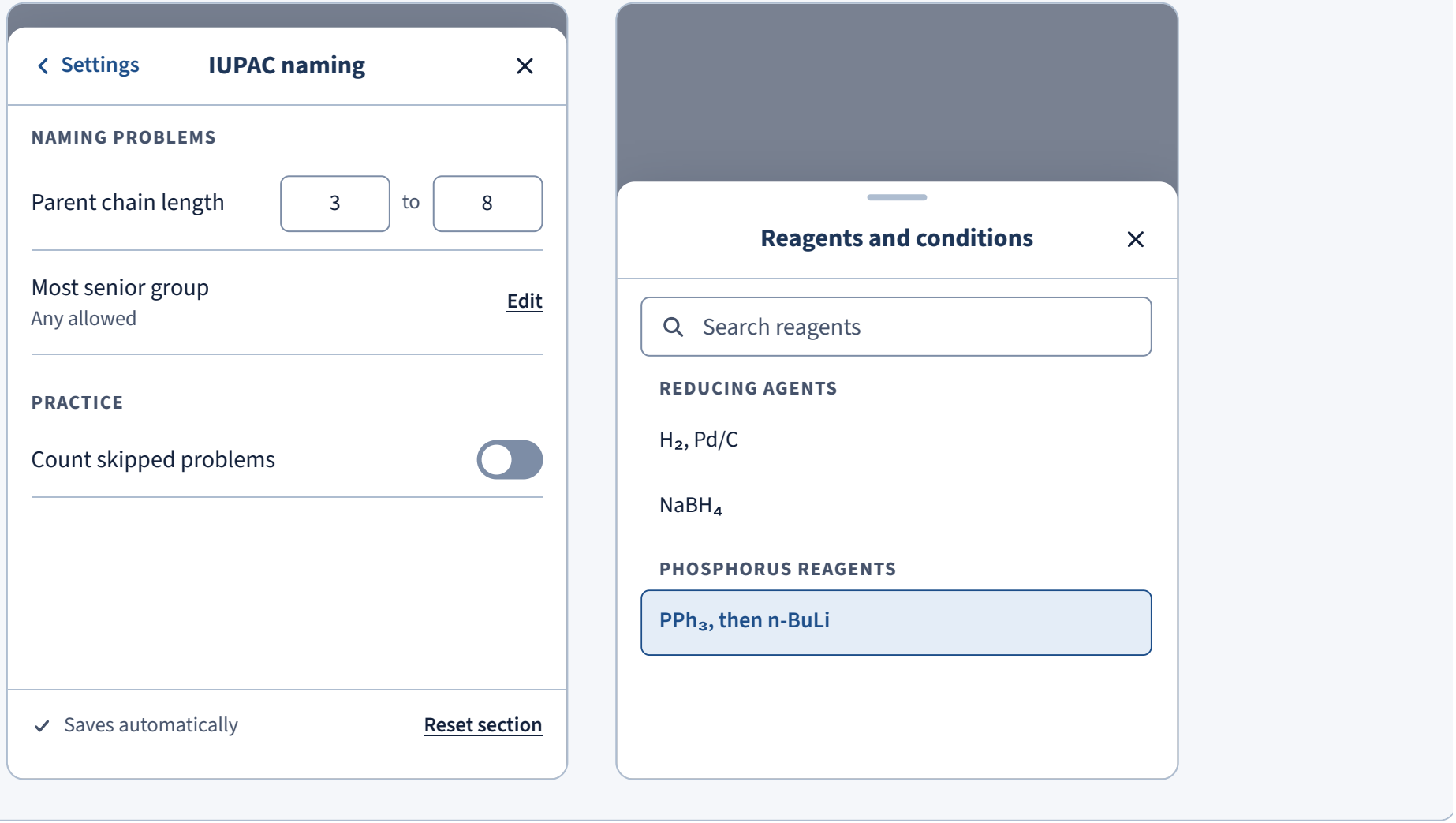


The one settings window for every generator: a wide Dialog with a section list on the left (Molecules, then each generator) and that section's settings on the right.

- Markup: `<div class="oh-dialog oh-settings" role="dialog" aria-modal="true" aria-label="Settings">` holding `<nav class="oh-settings__nav">` and `<div class="oh-settings__main">` (an `oh-close`, then `oh-dialog__body`, then `oh-dialog__foot`). Width `dialog-width`, section list `dialog-nav-width`.

- Section list: `oh-navgroup` headings ("All generators", "Generators") in `overline`, and an `oh-navitem` per section. The open section has `aria-current="page"` (`brand-tint` ground, `brand` text and outline). A coming-soon generator is `oh-navitem--soon` with a small "Coming soon" Badge. A section with invalid values shows an `oh-alert-dot` ("!", `error`) at the right.
- Body: a `title`-less heading (`oh-dialog__title`), an `oh-dialog__lede` line, then SettingRows grouped under section labels.
- Footer: `oh-saved` ("✓ Changes save automatically and apply to your next problem") left, "Reset this section to defaults" `oh-link` right. There is no Save button. On a first visit the footer instead holds "Use defaults" (link) and a primary "Start practicing" button.
- Opens from the ⚙ Settings button on a generator page straight to that generator's section; from the Account menu it opens to Molecules. The page behind keeps its place.
- Tablets: same window, nearly full screen, section list about 200px. Phones: the Sheet, with the section list as its first page.

Sheet



The phone container for anything that is a window or side list on laptops: settings (full height) and pickers such as the disconnection or reagent list (from the bottom).

- Markup: an `oh-scrim`, then `<div class="oh-sheet" role="dialog" aria-modal="true" aria-labelledby="…">`; add `oh-sheet--full` for settings. Parts: `oh-sheet__grab` (the drag handle, bottom sheets only), `oh-sheet__bar` (left: an `oh-back` "‹ Settings" button or nothing; centre: `oh-sheet__title`; right: `oh-close`), `oh-sheet__body` (scrolls), optional `oh-sheet__foot`.

- Rows inside are at least `touch-min` (44px). Navigation rows in a sheet become full-width list rows with a `›`.

- Settings: the first page is the section list; tapping a section slides in its page with "‹ Settings". Same rows and validation as the laptop window; an invalid section shows the `oh-alert-dot` in the title and the list.
- Close with ✕, a swipe down or the back gesture.

Tabs

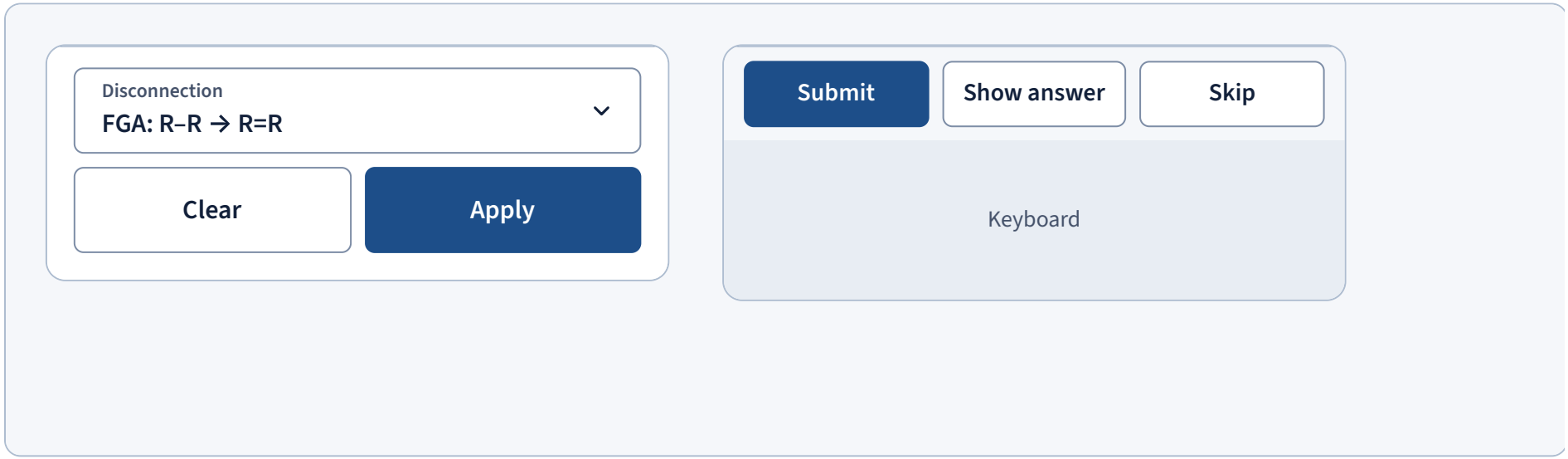


Two or three views of the same task inside a Dialog ("Draw" and "Enter SMILES").

- Markup: `<div class="oh-tabs" role="tablist">` with `<button class="oh-tab" role="tab" aria-selected="true|false" aria-controls="…">`; each panel is `role="tabpanel"`. Arrow keys move between tabs.
- Selected: `navy` label with a 3px `brand` underline; others `ink-muted`. Labels are short, verb or noun, sentence case.
- Not for page navigation: the site's sections use the Header nav and Breadcrumb.

Components: Generators

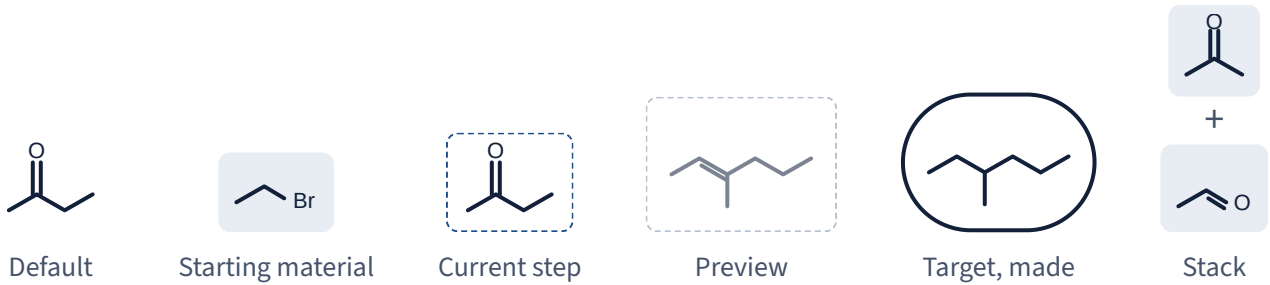
ActionBar



The phone bar that keeps a generator's actions in thumb reach: pinned to the bottom of the screen, or riding on top of the keyboard while the student types.

- **Pinned** (`oh-actionbar`): `surface-raised` with a `mist` rule above, sticky at the bottom, safe-area padding. Holds full-width buttons: the primary first ("Next problem", "Apply"), secondary pairs in `oh-actionbar__pair` ("Clear" · "Apply"), or a Picker above the buttons.
- **On the keyboard** (`oh-actionbar--keyboard`): Submit, Show answer and Skip in one row of 40px buttons, so the keyboard never has to close to answer. The keyboard's return key also reads "Submit".
- Laptops and tablets don't use it: their actions sit in the page.

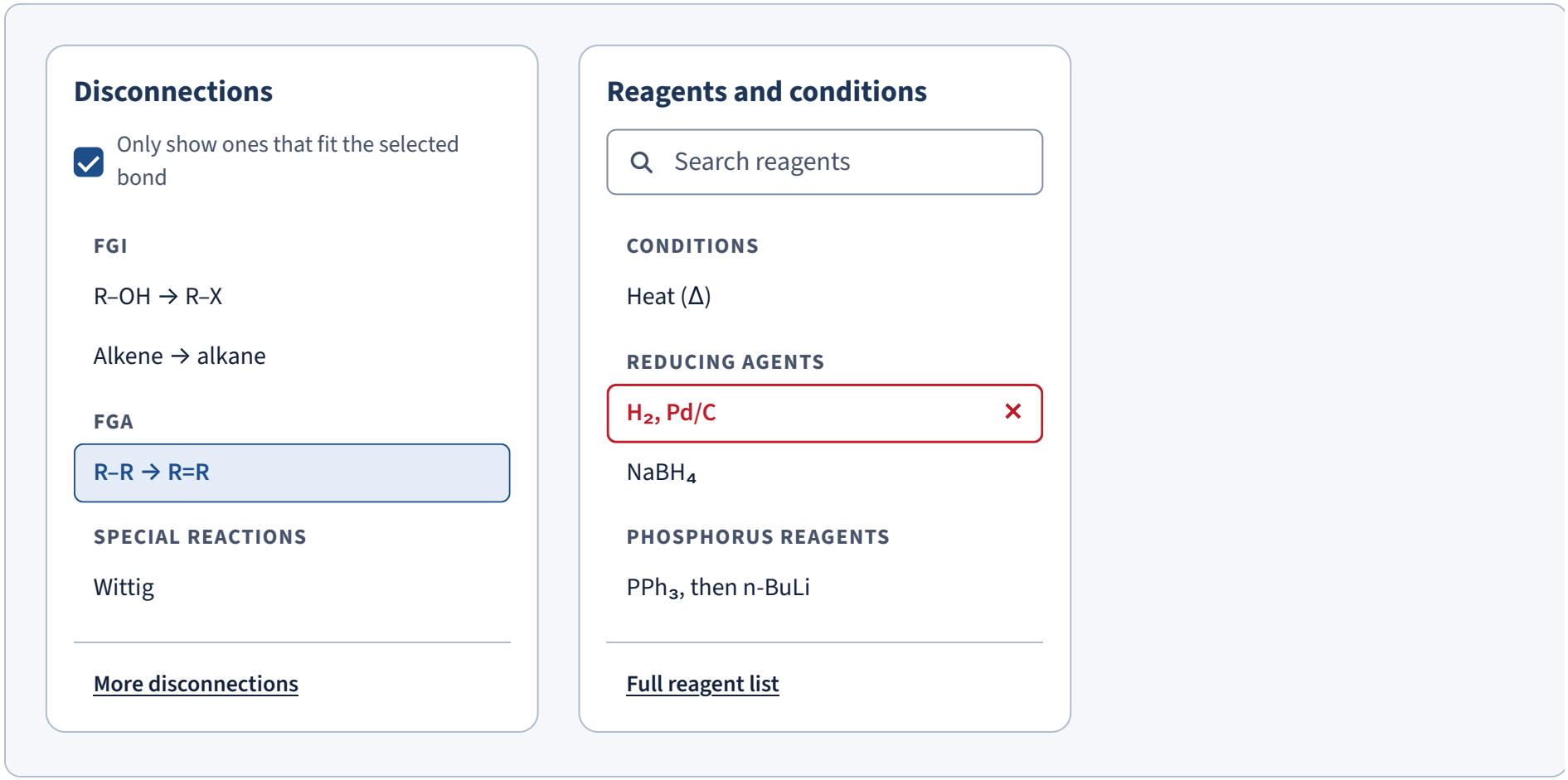
MoleculeChip



A small molecule drawing that stands for one step's molecule in the plan, a strip, a choice or the workspace, with a state shown by its frame.

- Markup: `<span class="oh-mol">` (or `<button class="oh-mol">` when tapping it reloads that step) holding the drawing in `navy` line work. Several reactants stack with `oh-mol-stack` (vertical) or `oh-mol-pair` (side by side) and an `oh-plus`.
- States: **starting material** `oh-mol--start` (`surface-sunken` ground: it can be bought, so it doesn't need to be made); **current step** `oh-mol--current` (dashed `brand` frame); **target** `oh-mol--target` (circled in `navy`, at the end of a finished forward synthesis); **preview** `oh-mol--preview` (dashed, faded: the result of a disconnection not yet applied, or a product not yet earned).
- Every drawing has an `aria-label` with the molecule's name, and a legend explains the shading where it first appears ("Shaded molecules are starting materials").
- The drawing style itself is still a placeholder (see MoleculePanel); only the frames are part of the system.

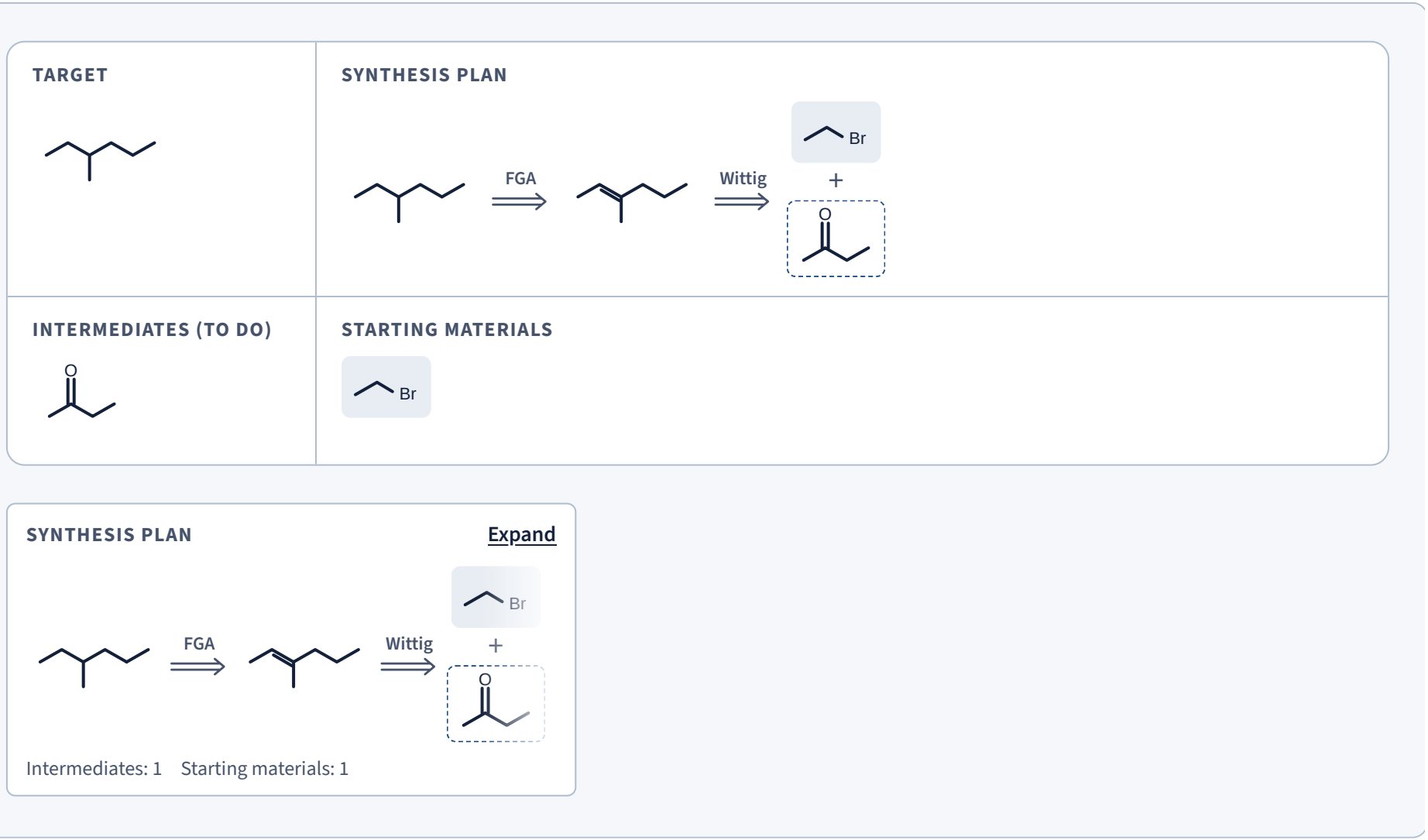
OptionList



The side list beside a generator's workspace: every disconnection, or every reagent and condition, grouped under small headings; the student picks one.

- Markup: `<div class="oh-optlist">` with an `oh-optlist__title`, optional `oh-optlist__tools` (a Search field and/or a filter Checkbox: "Only show ones that fit the selected bond"), then `role="listbox"` holding `oh-optlist__group` headings (`overline`) and `<button class="oh-option" role="option" aria-selected>` rows. End with `oh-optlist__more` for a "More disconnections" or "Full reagent list" `oh-link`.
- Selected: `brand-tint`, `brand` text and outline. A reagent that was tried and is wrong: `oh-option--wrong` (`error` text and outline, with a ✕), until the next pick.
- Sits in the `oh-gen` layout, `sidebar-width` wide. Long lists (the full reagent list) always get Search at the top.
- **Phones**: the list moves into a bottom Sheet, opened from a Picker above the actions; rows grow to `touch-min`.

PlanOverview



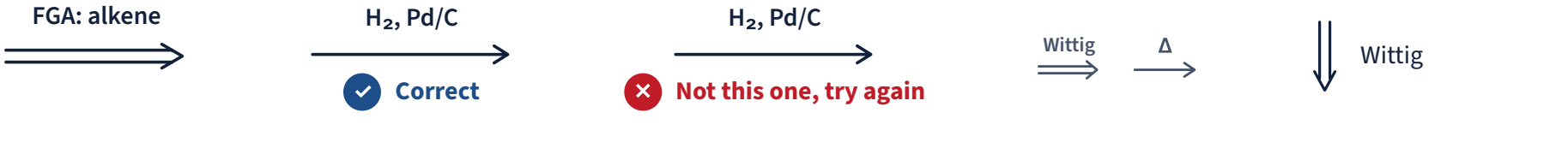
The synthesis problem at a glance, above the workspace: the target, the plan so far, intermediates still to break down, starting materials and, in stage 2, the forward synthesis.

- Markup: `<div class="oh-plan">` holding `oh-plan__cell`s, each with an `overline` label and an `oh-plan__row` of MoleculeChips and small ReactionArrows. Target sits in the narrow left column beside Synthesis plan; Intermediates beside Starting materials; Forward synthesis (`oh-plan__cell--full`) spans the bottom. Mark the bottom row `oh-plan__cell--last`.

- The plan is a mini-map: each molecule is a `button.oh-mol` that reloads that step; the step being worked on is `oh-mol--current`.
- An empty cell says so in `oh-plan__empty` ("None yet").

- **Phones** (`oh-plan-strip`): one horizontally scrolling strip (fading at the right edge) with an "Expand" link and counts underneath; Expand opens the full plan as a vertical tree in a Sheet.

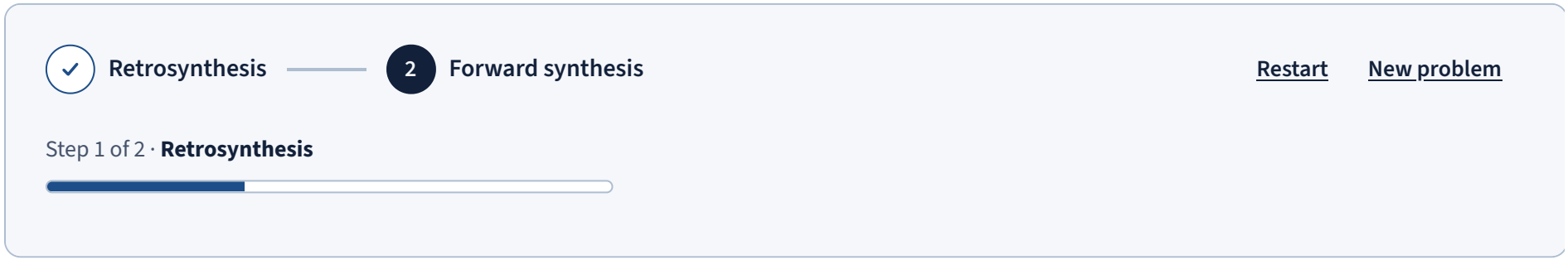
ReactionArrow



The arrow between molecules: a double-line open arrow (⇒) for a retrosynthetic step, a single arrow (→) for a forward reaction, with its label and, in forward synthesis, the verdict.

- Markup: `<span class="oh-arrow"><span class="oh-arrow__above">Wittig</span><svg…/><span class="oh-arrow__below">…</span></span>`. Above: the disconnection name (retro) or the chosen reagent (forward, semibold). Below: a Verdict once graded.
- **Verdict** (`oh-verdict`): ✓ "Correct" in `success` , or `oh-verdict--wrong` ✕ "Not this one, try again" in `error` . Words always accompany the icon. A wrong reagent leaves Next step disabled and the product as a preview.
- Sizes: default (workspace, 110–120px), `oh-arrow--sm` (inside PlanOverview and strips), `oh-arrow--down` (vertical, on phones, label beside the arrow).
- Arrow line work is `navy` , 1.8px; small arrows 1.4px.

Stepper



The two stages of a synthesis problem (Retrosynthesis, then Forward synthesis), with the problem's Restart and New problem links at the right.

- Markup: `<ol class="oh-stepper">` with `<li class="oh-step">` per stage holding `<span class="oh-step__num">1</span>` and the stage name, `oh-stepper__line` between, and `oh-stepper__end` for the links.
- Current stage: `aria-current="step"`, number filled navy. Done: `oh-step--done`, number replaced by a ✓ in brand. Later: ink-muted.
- **Phones** (`oh-stepper-compact`): one line "Step 1 of 2 · **Retrosynthesis**" over a thin Progress track; Restart and New problem move into the title's ⋯ menu.